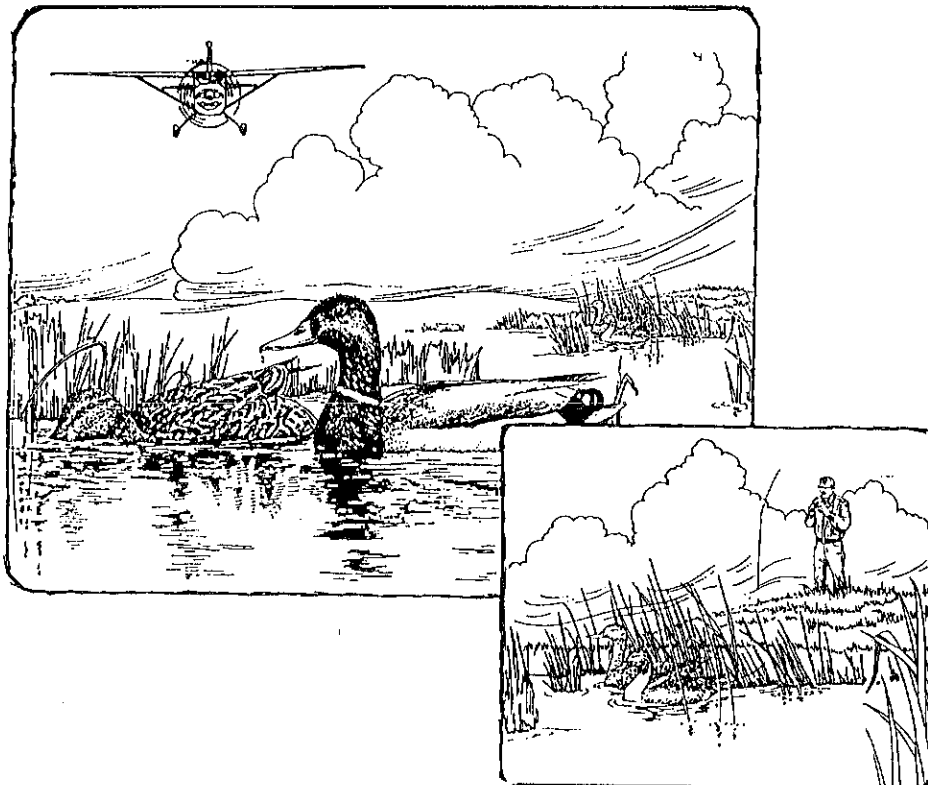


Enclosed is the current Standard Operating Procedures Manual for Aerial Waterfowl Breeding Ground Population and Habitat Surveys in North America. This manual replaces a similar document, dated April 1977. The basic operation of these surveys remains the same. However, some aspects of the survey program have changed in the last 10 years and other modifications will likely occur in the near future. These changes are the result of a constant review process designed to maintain the usefulness and integrity of this important data gathering effort. Future revisions to the manual will be mailed to you and should be incorporated into the looseleaf format where appropriate.

If there are additional requests for manuals, please contact:

WPS Section
Branch of Surveys
Office of Migratory Bird Management
Patuxent Wildlife Research Center
Laurel, MD 20708

Standard Operating Procedures
for
AERIAL WATERFOWL BREEDING GROUND POPULATION AND HABITAT SURVEYS
IN NORTH AMERICA



U.S. Department
of the Interior
Fish and Wildlife
Service



Environment
Canada
Canadian
Wildlife Service



April 1987

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SECTION I: INTRODUCTION

The Migratory Bird Treaty Act of 1918 (as amended) implements treaties between the United States of America and Great Britain (Canada, 1916), the United Mexican States (1936 as amended in 1972), Japan (1972 as amended in 1974), and the Soviet Union (1978). Under this Act, the Secretary of the Interior (United States) and the Minister of Environment (Canada) are responsible for the protection of migratory bird species.

In facilitating this task, an essential element has been the monitoring of various species population levels. In the late 1940's, the U.S. Fish and Wildlife Service developed aerial waterfowl breeding population and production surveys. These surveys were designed to provide annual breeding population and production indices covering changes over a major portion of the duck breeding range in North America. Today, these surveys continue to provide a basis for the management of most duck species on this continent.

The aerial Waterfowl Breeding Population and Habitat Survey was initiated experimentally in 1947 but was not fully operational until 1955. Similarly, the aerial Waterfowl Production and Habitat Survey was established experimentally in 1950 and became operational in 1956. Since their inception, these surveys have been conducted annually and without interruption. In 1959, air-ground comparison segments were started throughout most of the prairie crew areas to provide visibility correction factors for each species. The correction factors are applied to data obtained on the Waterfowl Breeding Population and Habitat Survey. These air-ground segments did not become operational until 1961. An attempt at developing air-ground comparison segments for the Waterfowl Production and Habitat Survey was initiated in 1950 but, because of many inherent problems, this work was discontinued in 1967. Consequently, the Waterfowl Production and Habitat Survey results are not adjusted for visibility bias.

Currently, the federal- and state-conducted surveys (utilizing the techniques outlined in this manual) monitor waterfowl population and habitat changes over nearly 1.4 million square miles (3.6 million km²) of breeding habitat within portions of Alaska, the Yukon and Northwest territories, 5 western Canadian provinces, and 5 north-central states (Fig. 1). This area has been divided geographically into 9 federal and 2 state crew areas, each of which is surveyed by independent crews. Due to their general cover types, federal crew areas encompassing survey

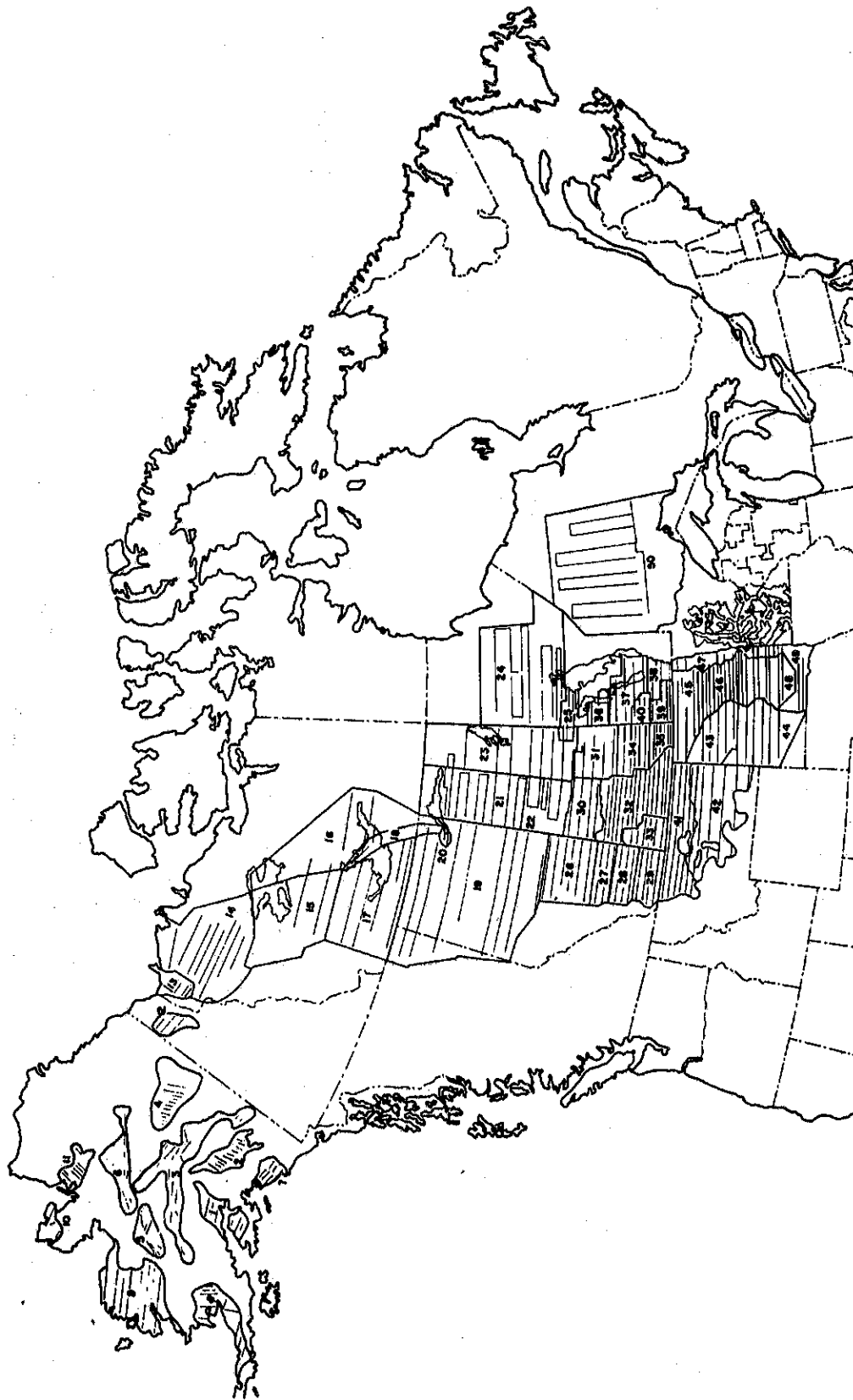


Figure 1. Survey strata and transects for principal areas of Waterfowl Breeding Population and Habitat Surveys in North America.

strata 1 through 25 and stratum 50 have been termed "bush" crew areas, whereas those encompassing strata 26 through 49 have been termed "prairie" crew areas. About 44,200 miles (71,100 km) of transects are covered during the Waterfowl Breeding Population and Habitat Survey which provides about a 1 percent sample of the combined strata areas. The Waterfowl Production and Habitat Survey covers roughly 30,700 miles (49,400 km) of transects to provide about a 0.5 percent sample of 0.8 million square miles (2.1 million km²) covered by this survey (Fig. 2).

The breeding ground surveys in Canada are a cooperative effort between the U.S. Fish and Wildlife Service and the Canadian Wildlife Service with the assistance of various other federal, state, provincial, and territorial resource management agencies. The standard operating procedures presented herein represent the latest refinement of survey methodology (Bowden 1973). Earlier contributors to these procedures are listed in the Bibliography. First-time readers of this document are cautioned to be aware that although there are great similarities between the Waterfowl Breeding Population and Habitat Survey and the Waterfowl Production and Habitat Survey, the methods and reporting procedures are described completely and separately without reference to each other.

The following Standard Operating Procedures Manual (henceforth referred to as the Waterfowl Breeding Ground Surveys Manual) has been prepared to standardize the procedures used in conducting these surveys. Involving more than 40 aerial and ground crew members in the federal survey units, these surveys require complex coordination effort. Although the procedures described here were formulated and finalized prior to 1955, minor revisions have become necessary with subsequent improvements in equipment and capabilities, organizational name changes, updating of long-term visibility rates, changes in survey coverage, etc. Therefore, this manual has been set up in loose-leaf form to facilitate periodic updates.

This revision supersedes the Standard Operating Procedures for Waterfowl Breeding Ground Population and Habitat Surveys, 1977. Copies of this edition are available on request from the Department of the Interior, U.S. Fish and Wildlife Service, Migratory Bird Management Office, Patuxent Wildlife Research Center, Laurel, Maryland 20708, and the Department of the Environment, Canadian Wildlife Service, Prairie Migratory Bird Research Centre, University of Saskatchewan, Saskatoon, Saskatchewan S7N 0X4.

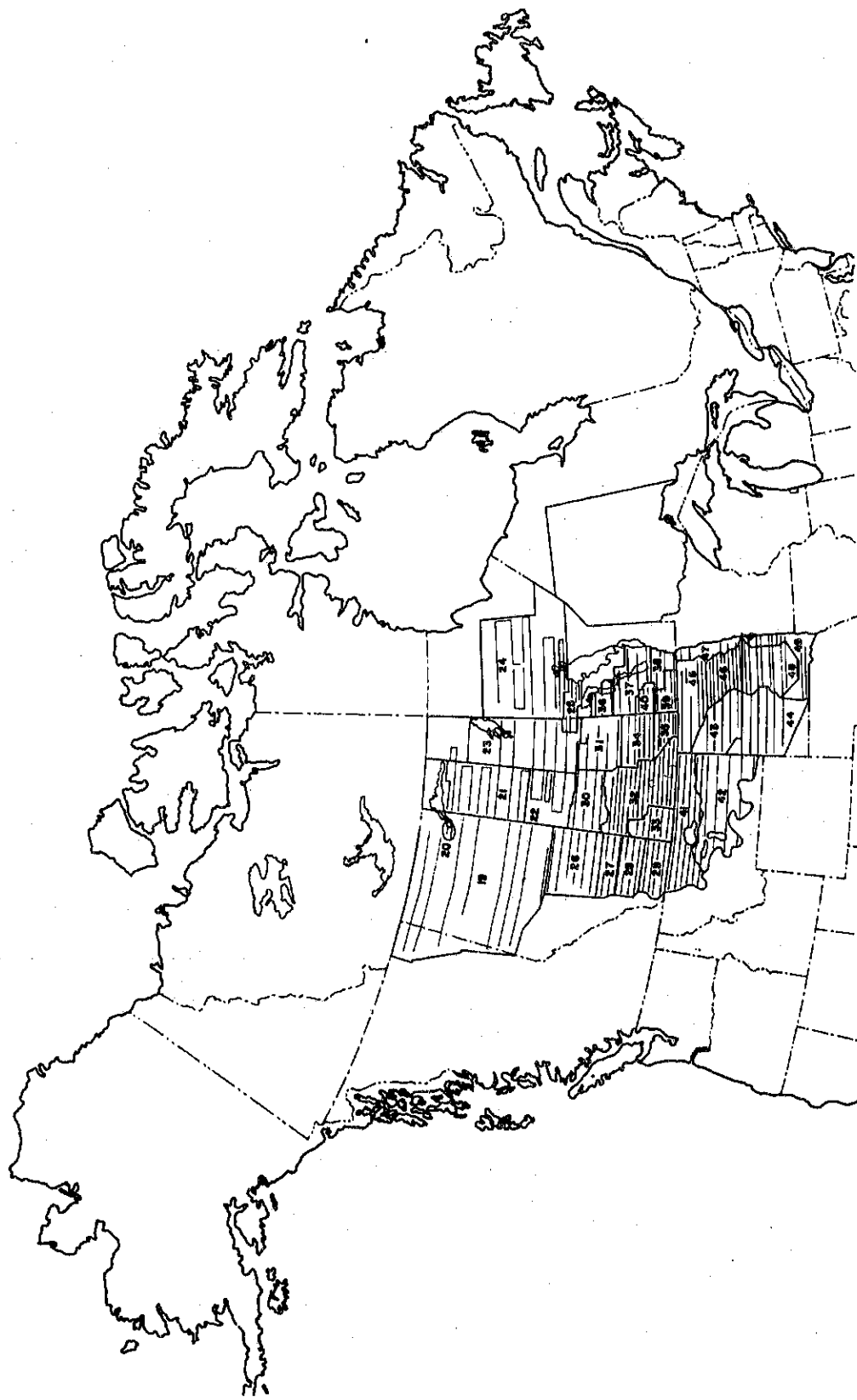


Figure 2. Survey strata and transects for principal areas of Waterfowl Production and Habitat Survey in North America.

SECTION II: RESPONSIBILITIES OF SURVEY PARTICIPANTS

A. U.S. Fish and Wildlife Service

1. Branch Chief-Surveys: The Branch Chief, Surveys, Migratory Bird Management Office (MBMO), United States Fish and Wildlife Service (FWS) is responsible for the compilation of survey data from all reporting units. This includes preparing the final waterfowl breeding population and production indices for the survey areas. These data become major components in the annual Fall Flight Forecast and Waterfowl Status Reports prepared by the FWS.

2. Section Chief - Waterfowl Population Surveys: The Section Chief-Waterfowl Population Surveys (WPS) is responsible for coordination of U.S. and Canadian personnel and insures the orderly conduct of the program in Canada and the United States. The Section Chief supervises all FWS personnel working on air and ground surveys in both the United States and Canada, and represents the FWS in all matters relating to the Canadian Wildlife Service (CWS), other Canadian officials, the Canadian public, and similar United States entities. The Section Chief is ultimately responsible to the Branch Chief-Surveys, FWS in reporting survey data.

3. Flyway Biologist: The Flyway Biologist is responsible to the Section Chief-WPS for the actual conduct of the waterfowl breeding ground population and habitat surveys in the assigned crew area. In this assigned crew area, the Flyway Biologist is in charge of the aerial and ground survey crews and coordinates the timing and sequence of ground surveys. The Flyway Biologist is the pilot-observer in the survey aircraft. Upon arrival in the assigned crew area, the Flyway Biologist contacts the appropriate state, provincial or territorial wildlife management agencies to inform them of planned activities and to offer an exchange of information. The Flyway Biologist, in cooperation with the Population Management Biologist (CWS) within the prairie provinces of Canada, compiles and analyzes the breeding population and habitat estimates to be included in the Waterfowl Breeding Population and Habitat Survey Report. The Flyway Biologist prepares and distributes the following reports to reach the addressees listed in the appendix by the established dates.

- a. Waterfowl Breeding Population and Habitat Survey Report
- b. Mid-May Habitat Conditions Report (submit only to Branch Chief-Surveys, FWS)
- c. Waterfowl Production and Habitat Survey Report
- d. Mid-July Habitat Conditions Report (submit only to Branch Chief-Surveys, FWS).

Flyway Biologists conduct surveys in specified portions of the Northwest Territories, British Columbia, Alberta, Saskatchewan, Manitoba, Ontario, Montana, North Dakota, and South Dakota.

4. Waterfowl Biologist-Waterfowl Investigations: The Waterfowl Biologist-Waterfowl Investigations stationed in Alaska conducts the surveys in specified portions of Alaska and the Yukon Territory. This individual performs the same duties outlined for Flyway Biologists and provides survey results (including photography) to the Assistant Branch Chief-Surveys, FWS by the required due dates.

B. Canadian Wildlife Service

1. Head-Population Management Section: The Head-Population Management Section (PMS), Western and Northern Region, is responsible for the coordination of activities between the 3 Canadian air-ground comparison survey crews (i.e. participation requirements, consistency of ground crew participants, procedural discrepancies, etc.), and the orderly conduct of the ground crew activities in Canada.

2. Population Management Biologist: The Population Management Biologist is responsible to the Head-PMS, for coordination of activities of the air-ground comparison survey crew, consistency in the adherence to survey procedures, and the compilation and analysis of ground survey data in the assigned area. In cooperation with the Flyway Biologist, FWS, this individual assists in the analysis of breeding population and habitat estimates to be included in the Waterfowl Breeding Population and Habitat Survey Report. The Population Management Biologist prepares and submits the Provincial Waterfowl Status Report in sufficient time to reach the Head-PMS by June 10.

C. State Wildlife Management Agencies

Waterfowl Biologist: The states of Minnesota and Wisconsin have participated in these cooperative surveys using comparable procedures since 1964 and 1973, respectively. Waterfowl Biologists for these states are responsible for surveys in specified portions of their respective states. These individuals perform the same duties as outlined for Flyway Biologists and provide survey results and appropriately labeled color slides as indicated to the Assistant Branch Chief-Surveys, FWS by the required due dates.

SECTION III: WATERFOWL BREEDING POPULATION AND HABITAT SURVEY

A. Methods

1. Survey Dates: The survey is performed in the prairie crew areas during the approximate period of 1 to 25 May. Differences exist in the chronology of the breeding season among years. In some years, most waterfowl en route farther north have departed the southern prairies by 1 May, and late-arriving waterfowl (e.g. blue-winged teal) have arrived and are occupying breeding territories. In other years, transient waterfowl in the southern prairies (e.g. scaup and certain other divers) may still be present in large mixed aggregations and blue-winged teal may not have arrived by 10 May. Therefore, to obtain the most representative breeding population information for the greatest number of duck species, the Breeding Population and Habitat Survey should not begin until the majority of more northern breeding species have moved through, and the bulk of late-nesting species have arrived and are dispersed into breeding territories. The survey should be initiated before early-nesting mallard and pintail drakes are commonly observed in post-breeding flocks of 3 and 4. In prairie Canada, the aerial crews coordinate closely with the CWS ground personnel to determine roughly when to plan crew arrival. Reconnaissance flights in each prairie crew area are conducted to determine the survey starting time.

The Breeding Population and Habitat Survey in the bush crew areas is performed during the approximate period 12 May to 12 June. Primary criteria used in bush crew areas for initiation of this survey are spring breakup, the build-up of representative waterfowl in traditional breeding areas, and the dispersal of paired waterfowl into breeding territories. As in the prairies, reconnaissance flights are made to determine the appropriate time for beginning the survey. The survey in Alaska begins about 18 May, depending entirely upon spring breakup.

2. Survey Flight Time: Daily survey flights begin no later than 1 hour after sunrise in prairie habitat and no later than 2 hours after sunrise in the parkland and southern bush habitats. The later start in parkland and bush areas allows for better light penetration into dense woodlands to improve visibility. In northern bush

habitats, survey flights are conducted during the mid-day period due to the extended daylight in northern latitudes at this time of year. Transect flying is completed by 12:00 noon (local standard time) each day in the prairies. Because of extended daylight and logistical problems associated with subarctic conditions, the completion time for bush operations is somewhat later in the day.

3. Survey Flight Conditions: Aerial surveys are not initiated when winds consistently exceed 15 mph (13 kts), if adverse weather conditions exist (i.e. snow, moderate to heavy rain, excessively rough air), or if visibility is poor for other reasons. Surveys are discontinued when winds exceed 25 mph (22 kts), turbulence is excessive, or if other adverse weather develops which is unsafe for flight or may compromise survey results.

4. Survey Flight Speed and Altitude: Transects are flown at ground speeds of 90-105 mph (78-90 kts). A stop watch is used to monitor aircraft ground speed and to aid in locating segment end points. Aircraft are normally flown at 100-150 feet (30-50 m) above ground level (AGL) to ensure accurate identification of waterfowl species under average light conditions and to assure safe obstacle clearance. Established known-width sample transects are flown at regular intervals to maintain consistency and accuracy of the aerial crew's transect width. Reference marks are applied to the aircraft's wing struts as guides for both pilot-observer and observer to indicate the approximate transect boundaries from the appropriate altitude.

5. Survey Flight Path: Transects are centered on section lines in the prairies and along degree lines of latitude or longitude in the bush. In Alaska, transects are oriented generally perpendicular to river valleys and coastlines. The transects are located as depicted in Fig. 1. The flight path is altered only to avoid flying directly over towns, game farms, poultry, livestock, or persons. The aircraft's flight path is not altered to facilitate identification of waterfowl. Transects are flown in the same direction each year as much as practical to minimize variance associated with changes between years.

6. Survey Sample: A segment is an 18-mile (29-km) sampling unit, 16-mile (26-km) in the Alaska crew area, having a total width of 1/4 mile (400 m), i.e. 1/8 mile (200 m) on each side of the aircraft for the waterfowl

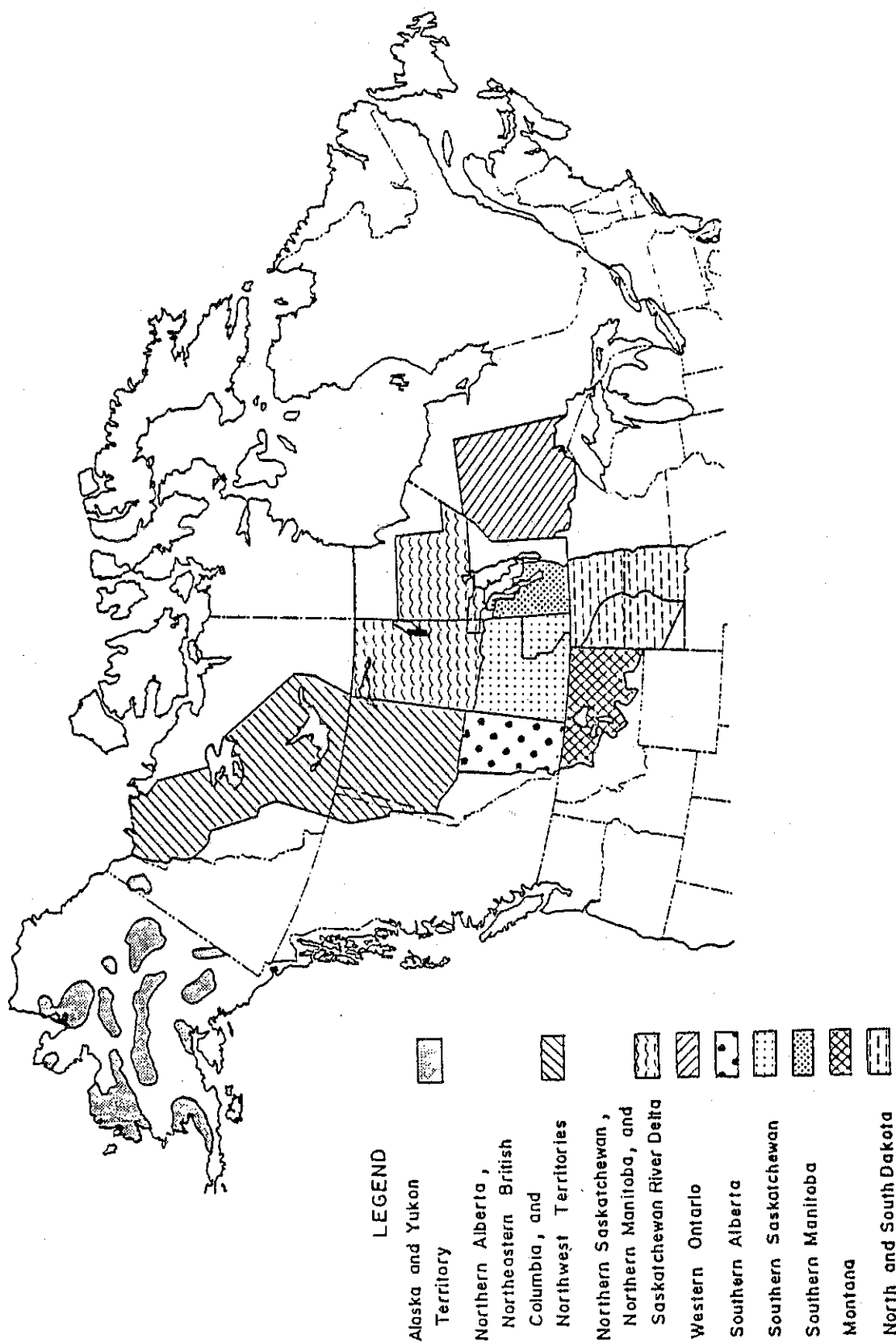


Figure 4. Reporting units for the federal aerial waterfowl breeding ground surveys.

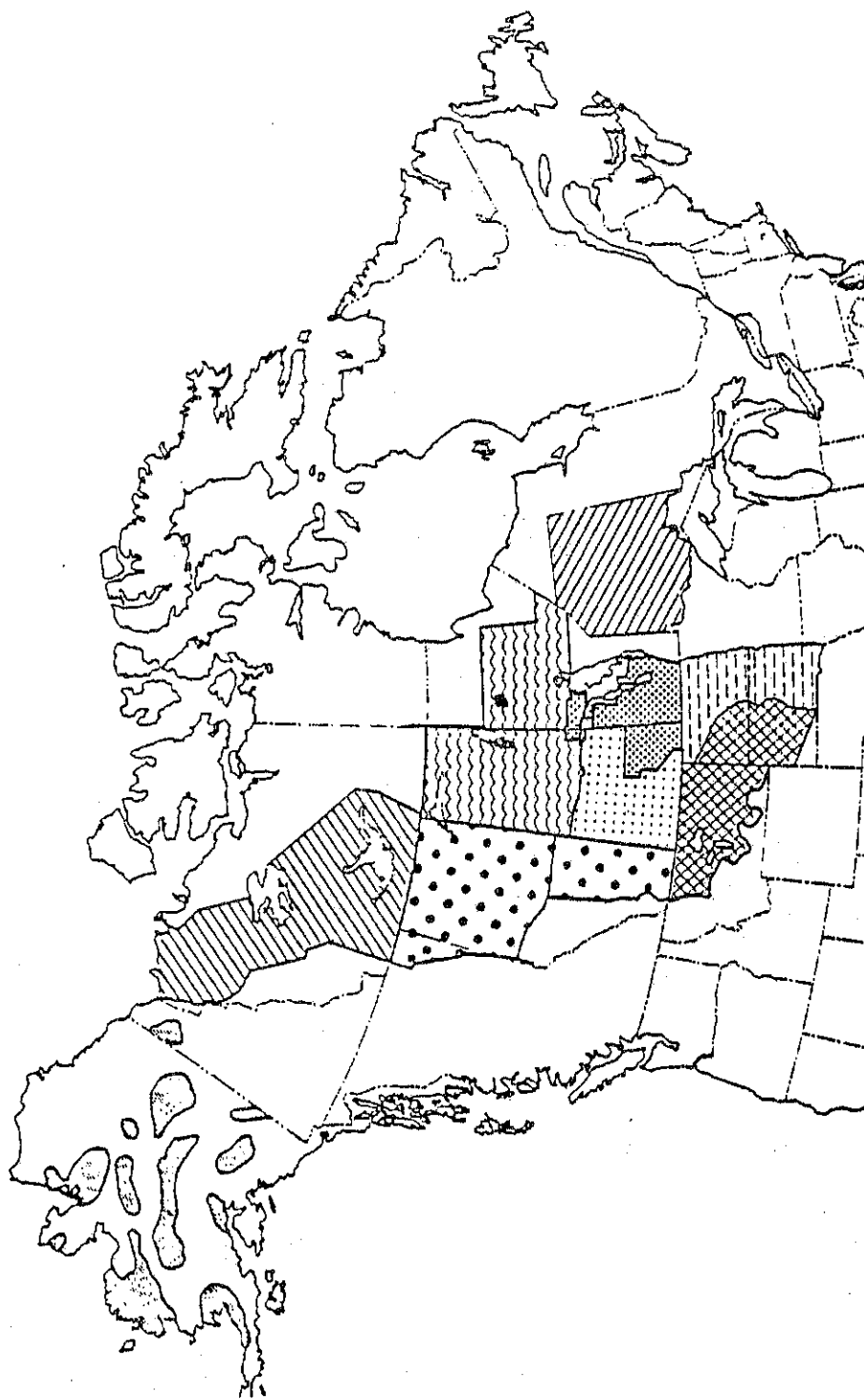


Figure 5. Crew areas for the federal aerial waterfowl breeding ground surveys.

Survey area	Crew Area			Reporting Unit		
	Strata	Area (Sq. mi.)	Sample (Sq. mi.)	Linear miles	Strata	Area (Sq. mi.) Sample (Sq. mi.)
<u>FEDERAL</u>						
<u>Bush</u>						
Alaska and Yukon Territory	1-12	81,770	912.0	3,648.0	1-12	81,770 912.0
Northern Alberta, Northeastern	13-18	273,201	927.0	3,708.0	13-20	438,687 1,669.5
British Columbia, and Northwest Territories						
Northern Saskatchewan, Northern	21-24	217,314	1,417.5	5,670.0	21-25	224,130 1,534.5
Manitoba, and the Saskatchewan						
River Delta						
Western Ontario	50	176,609	607.5	2,430.0	50	176,609 607.5
Bush subtotal		748,894	3,864.0	15,456.0		921,196 4,723.5
<u>Prairies</u>						
Southern Alberta	19,20,26-29	229,783	2,056.5	8,226.0	26-29	64,297 1,314.0
Southern Saskatchewan	30-33	88,912	967.5	3,870.0	30-35	111,120 1,269.0
Southern Manitoba	25,34-40	67,752	850.5	3,402.0	36-40	38,728 432.0
Montana	41-44	120,791	1,260.0	5,040.0	41-42	73,657 868.5
North and South Dakota	45-49	89,101	1,111.5	4,446.0	43-49	136,235 1,503.0
Prairie subtotal		596,339	6,246.0	24,984.0		424,037 5,386.5
FEDERAL total		1,345,233	10,110.0	40,440.0		1,345,233 10,110.0
<u>STATE</u>						
Minnesota		33,899	527.3	2,109.2		33,899 527.3
Wisconsin		43,359	412.5	1,650.0		43,359 412.5
STATE total		77,258	939.8	3,759.2		77,258 939.8
GRAND TOTAL (Federal and State)		1,422,491	11,049.8	44,199.2		1,422,491 11,049.8

Table 1. Waterfowl Breeding Population and Habitat Survey - crew area and reporting unit constants.

summary), and the original Segment Data Form are delivered to the Flyway Biologist responsible for that reporting unit as soon as possible following survey completion. Copies of the Field Data Forms and Field Data Form (stratum summaries) are retained by the aerial survey crew (aerial data) and the CWS ground crew leader (ground data) for referral should questions arise.

B. Data Collection

1. General: The following instructions are generally applicable to both the aerial and ground crews.

a. Strict adherence to the instructions and procedures set forth in this manual is required unless changes are approved by the Section Chief-WPS prior to the initiation of the surveys.

b. Survey data are recorded daily on the appropriate Field Data Forms.

c. Field Data Forms must be prepared in duplicate. It is not necessary to prepare duplicates of the Segment Data Forms (discussed later in this section). Copies are stored and mailed separately to reduce the chance of losing any field data.

d. The original field records are prepared neatly and legibly to facilitate auditing, duplication, and data transfer. Black ink or dark pencil (#2 or darker) is used on all forms.

e. Survey records are double-checked by the survey crew to ensure that transcriptions and mathematical calculations are accurate.

f. Instructions for completing the various forms are printed on the back side of each form and should be strictly followed.

g. The Segment Data Forms and originals of all aerial and ground Field Data Forms as well as stratum summaries are submitted directly to the Assistant Branch Chief-Surveys at the conclusion of the survey. Express Mail is used for this purpose whenever possible.

h. Color slides are useful in depicting habitat changes through the years, particularly as they

relate to availability of surface water and specific land-use activities detrimental to waterfowl. It is the responsibility of the ground crew leader to obtain and send photo material representing habitat change to the crew area Flyway Biologist; this material includes the unprocessed slides and a photo log with appropriate labels. These slides are then processed and labeled by the Flyway Biologist. It is the responsibility of the Flyway Biologist to obtain and appropriately label aerial color slides. These are forwarded with the ground crew slides to the Assistant Branch Chief-Surveys as soon as possible following survey completion, via Express Mail if available.

2. Waterfowl Data: Within the transect boundaries, all ducks (except lone hens) that can be identified by the aerial crews are recorded by segment on a tape recorder. When water areas occur on the boundary of a transect, only those waterfowl judged to be within 1/8 mile boundary on each side of the aircraft are counted. A separate recorder is used by each aerial crew member. These data are then transcribed to Field Data Forms each afternoon following the survey flights. (Exhibit 8-prairie crews and Exhibit 9-bush crews). The data are then summarized by stratum using the same forms (Exhibit 10-prairie crews and Exhibit 11-bush crews). It is emphasized that only those ducks within the transect boundaries on water bodies cut by the transect boundary are recorded. Unidentified ducks should not be recorded. Identified ducks in flight are recorded only if their flight originates or terminates within the transect boundary. For all species of ducks observed and identified, the number in each of the following categories is recorded. Instructions appear on the reverse of the Field Data Form.

- a. Lone drake--single isolated drake without a visible associated hen. (Lone hens are not recorded by the aerial crew.)
- b. Flocked drakes--2 or more drakes in close association (for the purposes of this manual, flocked drakes recorded on the Field Data Form are limited to groupings of 2 to 4 drakes).
- c. Pair--male and female in close association.

d. Group--3 or more of a mixed-sex grouping of the same species in close association which cannot be separated into singles and pairs (for the purposes of this manual, a hen and 2 drakes are recorded as a pair and a lone drake, and 5 or more flocked drakes are recorded as a group).

Coots are recorded as a total number for each observation. Geese observed on transect are recorded in the same manner as ducks.

It is emphasized that this is a breeding population survey and not a true breeding pair survey. The estimates derived include not only indicated pairs but also grouped birds, some of which are undoubtedly non-breeders and transient birds.

Table 1 (Exhibit 1) of the survey unit report contains a long-term trend in adjusted waterfowl breeding population estimates by species for the current year and all previous years of record. Table 2 (Exhibit 2) contains the current year's adjusted waterfowl breeding population estimates by species and stratum, showing comparisons of the current year with the previous year, the previous 10-year mean, and the long-term mean.

3. Water Data: Water data are collected only in the prairie and parkland habitats. The aerial observer records the number of individual water areas within 660 feet (200 m) on that side of the aircraft for each segment. The pilot-observer does not count ponds for reason of flight safety. Water areas intersected by the transect boundary are counted. A tape recorder is used for this purpose as is done for the waterfowl data. Two mechanical counters are often used to assist the observer in this task, particularly in high density pond areas. The number of natural water areas (e.g. ponds, lakes, rivers, streams) is recorded on one counter, while the number of artificial water areas (e.g. reservoirs, stock dams, dugouts, and large irrigation ditches) likely to contain water into mid-summer is recorded on the other. Natural and artificial water areas are identified as such to monitor human effects on natural wetlands over the history of the surveys. The number of ponds for each segment is the sum of the counter totals. Table 4 (Exhibit 4) contains the long-term trend in adjusted pond estimates by stratum, comparing the current year with the previous year, previous 10-year mean, and the long-term mean. Table 4 (Exhibit 4) represents types III, IV, and V wetlands (Shaw and Fredine 1956),

artificial areas, and streams only. Type I wetlands are tallied only on the ground crew field data and summary forms.

Natural water areas to be counted from the air include Type III, IV, and V wetlands, rivers, and streams. Type III ponds (seasonal wetlands) refer to hay meadows or basins containing natural aquatics which normally are dry by midsummer but are expected without additional precipitation to retain water for at least 3 weeks following the observation. These wetlands normally have a uniform vegetative cover and contain at least 6 inches (15 cm) of water. Under dry fall conditions, many ponds are plowed but refill with water in the spring. In these instances, no vegetation is present, but these wetlands should be counted if they have sufficient water depth and lasting qualities. To minimize subjectivity, it is stressed that water depth is the primary criterion. Type IV ponds (semi-permanent wetlands) have sufficient water depth that they will likely persist through the brood season but may become dry during late August or September. These wetlands usually contain water during at least 7 out of 10 years, and the vegetation is normally clumped, covering all but the center of the wetland. Type V ponds (permanent wetlands) are deep marshes or lakes having sufficient water depth to persist throughout the summer and fall. These wetlands normally are characterized by a peripheral rim of aquatic vegetation bordering an open body of water. Streams and rivers that meander through the transect are counted as separate water bodies each time they occur within the transect boundary. Dugouts that are inundated by a natural wetland are not recorded as artificial water areas, but if the dugout can be identified as a water area separately from the surrounding natural water area, both the dugout and the natural water area are counted.

The following water areas are not recorded by the aerial crew:

- a. Type I ponds (temporary wetlands) refer to temporary water, sheet water, small wet areas in stubble or plowed fields, and wet depressions that have less than 6 inches (15 cm) of water depth and can be expected to last less than 3 weeks.
- b. Roadside or borrow ditches where the water is confined entirely to the ditch and small ditches used for local irrigation.

c. Muskeg areas where water may glisten under a dense growth of grass. Open areas of water within muskeg are not counted.

4. Air-Ground Comparison Data: Aerial and ground crews conduct surveys of selected segments in each stratum to provide correction factors for visibility biases (waterfowl and ponds present but not observed by the aerial crew). Visibility biases are known to exist for all aerial observers. These biases change between years as a result of varying conditions (i.e. vegetative cover, cloud cover, light, wind, aircraft design and noise level, observer fatigue, observer experience, etc). These biases are not necessarily consistent between individual observers given the same set of circumstances. To minimize the effect of these factors, as many variables as possible are made consistent between years. Some variables cannot be controlled and must be dealt with annually. In the prairies, visibility rates for major species of ducks may vary significantly. Therefore, visibility rates are determined on an annual basis in each of the prairie crew areas. Although it is suspected that visibility rates for bush crews vary to a lesser degree, species visibility correction factors developed in prairie crew areas are applied to bush transects. (See Exhibit 12 for these visibility rates.) To develop more precise visibility correction factors applicable to bush crew areas, air-ground comparison segments are being established using helicopters.

a. Aerial Crew: The aerial crew covers each air-ground comparison segment in the same manner as other segments. Each air-ground segment is flown within a day of the time the immediately adjacent transects are flown (preferably on the same day). The aerial crew flies air-ground segments in early, mid-, and late-morning to obtain an even distribution of sampling effort. Likewise, the aerial crew distributes the number of eastbound and westbound air-ground segment flights proportionately to the transect flights flown within that stratum. The data are recorded on the Aerial Crew Field Data Form in the same manner that other segment data are recorded, but is prepared on a separate Field Data Form (Exhibit 13) from the operational transect data.

b. Ground Crew: The ground crew consists of no less than 2 and no more than 4 individuals. The

ground crew initiates segment coverage within 1-1/2 to 2 hours of sunrise. Every effort is made to complete the ground segment by 12:00 noon (local standard time). However, coverage will be completed within one day. Ground coverage of all air-ground segments is from east to west to minimize visibility problems associated with sun glare. The ground crew covers each segment on the same day or within 2 days following aerial coverage. Every effort is made to begin the segment on the morning of the day immediately following aerial coverage. Ground coverage is not initiated until at least 1/2 hour after aerial coverage has been completed if conducted the same day. Ground coverage is not initiated or continued when winds are in excess of 25 mph (40 kph) or if other adverse weather conditions exist (e.g. moderate to heavy snow, heavy rain, thick smoke, heavy fog). If the ground crew cannot complete coverage within the 3-day specified period, a second aerial count is required. The ground coverage, as before, is completed on the same day or within 2 days following aerial coverage. Landowners are contacted whenever possible prior to entry upon their land.

It is emphasized that the primary objective of the ground work is to get a complete count of all waterfowl present on the air-ground comparison segment. Whenever there is any doubt regarding the ground crew member's ability to see the entire surface of a water body and all waterfowl that might be resting on its shore, the ground crew member walks around the pothole to the extent necessary to ensure sighting and identifying all waterfowl. In instances where there are large potholes with open shorelines and large numbers of waterfowl, the ground crew member refrains from flushing the waterfowl since this may result in counting error due to "roll-up" (counting the same waterfowl at 2 or more locations) or inability to identify all the birds in flight due to poor positioning or dealing with too many birds at one time (mass flights). Waterfowl that are flushed will be recorded and watched to the extent necessary to rule out duplicate counting further along the segment.

When a pothole falls on the boundary of a segment only those waterfowl are counted that are judged to be within the segment at the time the ground

crew arrives at the wetland. Waterfowl in flight over the segment are counted only when they are obviously associated with the segment. Waterfowl are recorded in the same manner as by the aerial crew. However, all waterfowl present on the segment must be identified and included in the count. Unlike the aerial crew, the ground crew records certain lone hen data. Lone hen redhead, scaup, ring-necked duck, and ruddy duck are recorded. Lone hens of other species are not recorded (Bowden 1973). This is explained in the Data Adjustment Section to follow. To minimize "roll-up" and inability of the ground crew to identify all waterfowl during mass flights, firecrackers, shellcrackers, uncontrolled dogs, and other scare devices are not used.

All-terrain vehicle use is minimized for the same reason. However, limited use may be warranted in situations where access would not otherwise be possible. Direct inspection of ponds while on these vehicles is avoided. The noise produced by these vehicles masks the ability of ground crew members from cuing to ducks by sound. For reason of safety, all crews using all-terrain vehicles strictly adhere to prescribed operating guidelines.

The ground crew tabulates and classifies all water areas within the entire segment width of 1/4 mile (400 m). In addition to the kinds of water areas counted by the aerial crew, Type I wetlands are recorded by the ground crew. These temporary wetlands are sheet water, small wet areas in stubble or plowed fields, and wet depressions that are less than 6 inches (15 cm) deep and can be expected to last less than 3 weeks under normal conditions. Type I ponds are important to the ground survey for the purpose of monitoring specific habitat condition changes that cannot be readily measured by the aerial crew. In cases where ponds join because of high water levels, the ponds are recorded as one pond and are noted as such on the data forms.

Additionally, pond water level data can be gathered in a qualitative manner by the ground crew. A system for doing this was developed by Millar in 1980. For this system he defined a basin as the total physical depression capable of holding water and a wetland as the portion of the

basin which contains any wetland vegetation. He further defined specific water level as follows:

- Stage 1 DRY - surface water has disappeared completely due to seepage, evaporation, or drainage.
- Stage 2 VESTIGIAL - water occurs only as small pools or puddles within the central vegetative zone of the wetland and can be expected to dry up within a matter of days.
- Stage 3 RECESSIONAL - water levels have receded within the central zone but still cover a fairly extensive area. In Open Water type of wetlands there is a drawdown mudflat between the water edge and the emergent vegetation. In Emergent Deep marshes and Shallow marshes, a fairly wide band of the central zone is DRY.
- Stage 4 INTERMEDIATE - in Open Deep marshes and Shallow Open Water wetlands, the central open water zone is completely flooded and the water extends into the inner edge of the emergent vegetation. In Emergent Deep marshes and Shallow marshes, the water extends throughout most of the central zone.
- Stage 5 FULL - the wetland is filled to the outer limit of the wet meadow zone.
- Stage 6 FLOODED - water has spilled out of the wetland proper and floods the upland vegetative communities in the basin.
- Stage 7 OVERFLOWING - water is at full supply level (FSL) of the basin and is spilling over. FSL can vary greatly from basin to basin. In some basins it is at the level of the outer edge of the wet meadow zone and in others FSL is so high that it will never be attained (e.g. terminal wetlands).
- Stage 8 DUGOUT-RECESSIONAL - water has receded well below the rim of the dugout.

Stage 9 DUGOUT-FULL - water level is at or close to the rim of the dugout.

Pond and waterfowl data collected by the ground crew for each water area include the following:

(1) pond number--designated from potential basins when the transects were established.

(2) pond type (Shaw and Fredine 1956).

(3) water level (Millar 1980).

(4) margin width--the area extending up to 33 feet (10 m) beyond the outer edge of the wet meadow zone.

(5) type and percentage of impaction to wetland basin (burned - autumn or spring, cultivated, drained, filled, grazed, hayed, or impacted by construction)--the basin is defined as that area from the center of the wetland to the outer edge of the wet meadow zone.

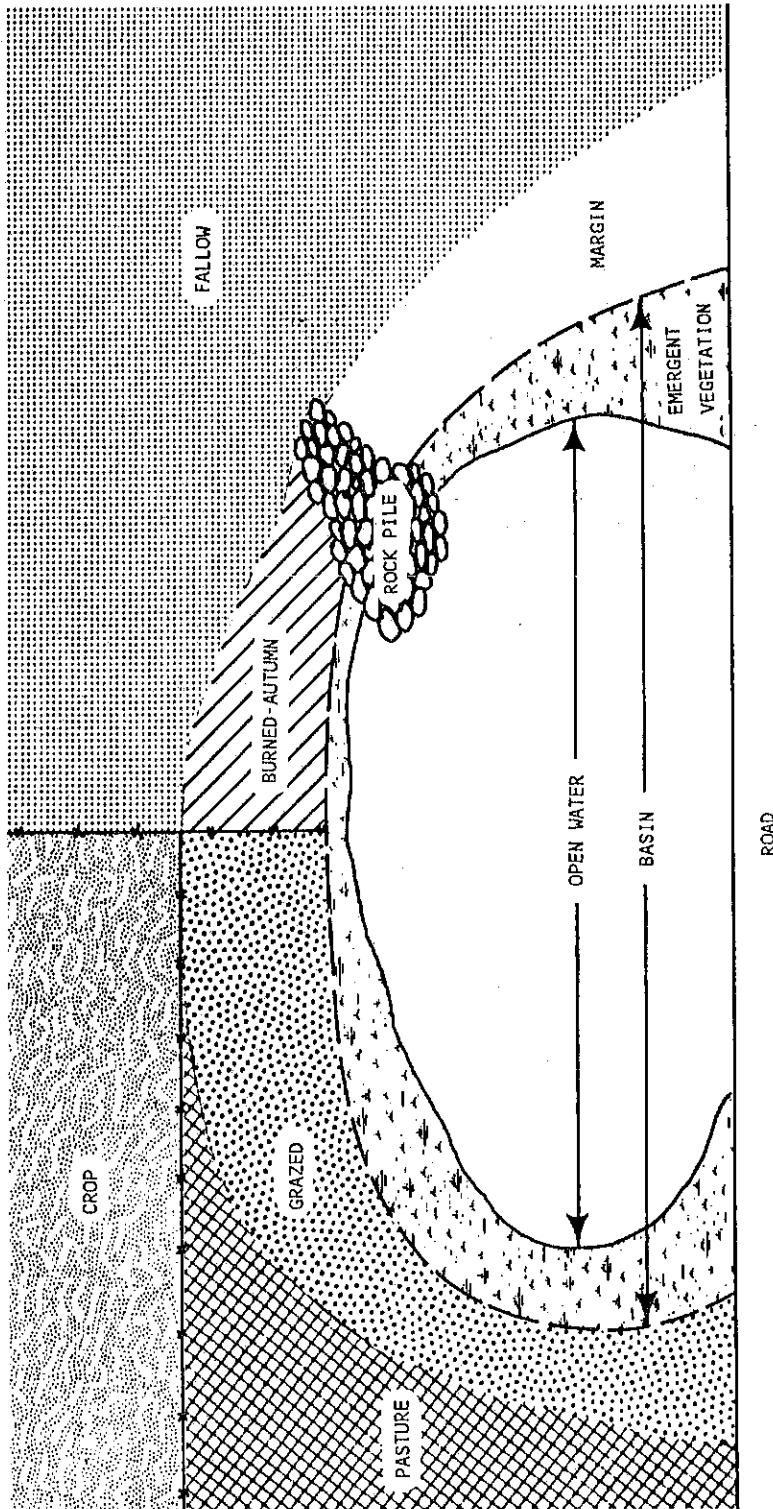
(6) type and percentage of impaction to wetland margin (burned - autumn or spring, cultivated, cleared, grazed, hayed, or impacted by construction).

(7) type and percentage of impaction to the uplands (crop, fallow, grassland, hayland, pasture, stubble, or woodland)--the upland is defined as the area contiguous with the margin.

Note: Fig. 6 provides a sample diagram of various types of wetland impacts as discussed in items 5, 6, and 7 above.

(8) the number of lone drakes, certain lone hens (redhead, scaups, ring-necked duck, and ruddy duck), flocked drakes, pairs, and grouped waterfowl of each species observed.

The starting and finishing time and certain weather data are recorded for each segment. These data are entered on the Air-Ground Comparison Segment--Ground Crew Field Data Form and the Ground Crew Summary Form (Exhibits 14 and 15).



Wetlands				
Pond No.	Pond Type	Water Level	Margin Width	Impact (%)
				Basin Margin Uplands
1	5	3	8	5F 50G 40P 20I 20BA 50E

Figure 6. Wetland impact diagram .

The original Field Data Forms and Summary Forms are delivered to the Flyway Biologist in charge of the reporting unit at the completion of the ground survey. These data become part of the Waterfowl Breeding Population and Habitat Survey Report for that reporting unit.

C. Data Adjustment

1. Prairies: Data collected by the ground and aerial survey crews are entered on the Air-Ground Visibility Rate Calculation Form (Exhibit 5) to calculate the visibility rate for each species. It is necessary to combine air-ground segment data from all strata within a crew area to develop reliable visibility rates for each specific crew area. This pooling of data is necessary to reduce the variance associated with small sample sizes. The following table lists all prairie crew areas and strata in the first and second columns and strata used for adjustment purposes in the third column:

Table 2. Strata used for adjustment of data by crew area on air-ground segments.

<u>Crew areas</u>	<u>For stratum</u>	<u>Combine air-ground data from strata</u>
<u>Prairie</u>		
Southern Alberta	26	26-29
	27	26-29
	28	26-29
	29	26-29
Southern Saskatchewan	30	30-33
	31	30-33
	32	30-33
	33	30-33
Southern Manitoba	34	34-40
	35	34-40
	36	34-40
	37	34-40
	38	34-40
	39	34-40
	40	34-40
Montana	41	41-44
	42	41-44
	43	41-44
	44	41-44
North and South Dakota	45	45-49
	46	45-49
	47	45-49
	48	45-49
	49	45-49

When the current year's calculated visibility rate for a particular species falls within the specified range (Exhibit 12), the current year's rate is used; when the current year's calculated visibility rate for a particular species falls outside the specified range (Exhibit 12), the long-term average crew area visibility rate is used in calculating the species index. In such cases, the substituted visibility rate is entered on the Air-Ground Visibility Rate Calculation Form as well as the Stratum Summary Form with appropriate footnotes. The long-term average species visibility correction factors developed in the combined prairie crew areas are always used for American black duck, goldeneyes, bufflehead, oldsquaw, eiders, scoters, and mergansers, as these species are rarely sampled in sufficient numbers during any given survey. Table 5 (Exhibit 5) of the survey unit report is the Air-Ground Visibility Rate Calculation Form and contains the current year's air-ground data along with any appropriately noted exceptions.

In preparing the Waterfowl Breeding Population and Habitat Stratum Summary Forms and the Air-Ground Comparison Segment--Ground Crew Summary Forms, lone and flocked drake redheads, scaups, ring-necked ducks, and ruddy ducks are multiplied by 1 to determine total indicated birds, rather than by doubling as is done for all other species. This procedure is used because these species are known to be late nesters and have significant disproportionate sex ratios leaning heavily towards males. Similarly, lone hens of these species observed by the ground crew are multiplied by 1 on the Ground Crew Summary Form.

In strata 34 through 40, ground crews have demonstrated that an positive relationship exists between mallard visibility rates and pond densities. To correct for this phenomenon, an adjusted mallard visibility rate is used in these strata. The method of calculating this rate was derived from work done by Bowden (1973) and is presented in Exhibit 16. This phenomenon has not been consistently demonstrated for any other species or in any other crew area.

2. Bush: As previously indicated, the average species visibility correction factors developed in the prairie crew areas over the history of the survey are applied in the bush crew areas. Work is continuing to develop more appropriate visibility rates in all bush units.

D. Data Analysis

Field Data Forms designed for the prairie crew areas and bush crew areas are used to tabulate waterfowl and water data for each segment, transect, and stratum. Data collected for each stratum are summarized on the Waterfowl Breeding Population and Habitat Survey Stratum Summary Form (Exhibit 17). Each stratum expansion factor is determined by dividing the total area (mi^2) in the stratum design by the total (mi^2) sampled within the stratum (current year). This factor is then used as a multiplier in obtaining the expanded waterfowl and pond indices. Waterfowl and pond visibility rates obtained from the Air-Ground Visibility Rate Calculation Form (Exhibit 5) are an integral part of the calculation on the Stratum Summary Form to obtain the adjusted waterfowl population and pond estimates. A copy of the Stratum Summary Form for each stratum becomes a part of the survey report and follows Table 7 (Exhibit 7).

In the prairies, a lone drake index (LDI) is used as a relative measure of the nesting season progress at the time of the survey. The survey is conducted at the most appropriate time when the LDI is highest for a given breeding season. A low LDI would indicate an early or late survey or even a poor nesting effort regardless of survey timing. A low LDI in conjunction with a detectable bird movement would suggest the survey was early. In contrast, a low LDI in conjunction with an unusually large number of post-breeding drake aggregations consisting of 4 or more birds would suggest the survey was late. Irregular weather patterns may complicate the interpretation. There are several species which do not exhibit synchronous nesting, regardless of the season; therefore, only certain species are used in deriving this index. Since the mallard, northern pintail, and canvasback normally begin to nest in early May and their numbers are comparatively high, the determination of the LDI is based on these species only. The method used to calculate lone drake indices is presented in Exhibit 18. Table 3 (Exhibit 3) of the survey unit report contains these data for the current and all previous years of record. These data are not tabulated for the bush units.

Flyway Biologists conducting surveys in Canada cooperatively analyze and interpret the field survey data with the Population Management Biologist, CWS, and

provide copies of all pertinent data before leaving Canada. The data are made available to the CWS as soon as possible after survey completion because of the importance of these data for the development of annual Canadian waterfowl regulations in early June.

E. Segment Data Form

The Waterfowl Breeding Population and Habitat Survey-Segment Data Form (Exhibit 19) is used to provide data for entry in the FWS primary computer. The form is completed following the tabulation of the data for each day's flight. Transcribing errors are greatly reduced by the observer and the pilot-observer auditing each other's work.

F. Survey Design

The survey design and the coverage of the current year's survey is presented in Table 7 (Exhibit 7). The first set of data presents the survey as it was designed by stratum including the area (mi^2) in each stratum, the area (mi^2) in the sample (both waterfowl and ponds), the number of linear miles in the sample, the number of transects in the sample, the number of segments in the sample, and the expansion factor. The second set of data is the current year's survey coverage. If the current survey is incomplete or has been modified, the second set of data will reflect these changes. The narrative section of the report explains the reasons for any omissions or changes.

G. Reports

1. Mid-May Habitat Conditions Report: During mid-survey, the Flyway Biologist in charge of each crew area contacts the Assistant Branch Chief-Surveys, by telephone, and provides a report regarding general habitat conditions including spring weather, and other events that "set the stage" for the survey since the previous summer (i.e. climatological, land-use changes), surface water abundance and relative water levels, soil moisture, timing of snow and ice disappearance, quality and timing of run-off, relative availability and quality of both upland and overwater nesting cover, phenology of vegetation, impact of agricultural practices, nesting chronology, overall production potential, and other pertinent information. This report is due on 15 May for prairie units and 25

May for bush units. A one-page report reflecting these observations is mailed at the time this information is called in. Attached to this is a roughed-out map of the survey unit including pertinent notes based on reconnaissance flights and survey observations made prior to the mid-survey reporting date. Express Mail is used for this purpose whenever possible.

2. Waterfowl Breeding Population and Habitat Survey Report: The Waterfowl Breeding Population and Habitat Survey Report is prepared using the following outline and suggested minimum text material.

COVER SHEET: The following statement is used--"The data presented in this report are preliminary and subject to further auditing. Final estimates will be available from the U.S. Fish and Wildlife Service, Migratory Bird Management Office, Patuxent Wildlife Research Center, Laurel, Maryland 20708."

TITLE: Waterfowl Breeding Population and Habitat Survey for (insert unit title)

STRATA SURVEYED: (insert strata numbers)

DATES: ____-____ May 19__

DATA SUPPLIED BY: United States Fish and Wildlife Service and Canadian Wildlife Service
(include CWS only for surveys conducted in southern Alberta, southern Saskatchewan, and southern Manitoba).

Air Crew

Pilot-Observer (insert name, affiliation, and title)

Observer (insert name, affiliation, and title)

Ground Crew

Crew Leader (insert name, affiliation, and title)

Assistants (insert names, affiliations, and titles)

ABSTRACT: A concise summary of weather and habitat conditions, the May pond estimates, lone drake indices, and breeding population estimates for the more numerous and important species is presented showing changes from the previous year, the previous 10-year mean, and the long-term mean. A brief statement is provided on the

overall waterfowl status in the reporting unit compared to previous years.

METHODS: A statement similar to the following begins this section of the report: "The procedures followed in conducting this survey are contained in the Standard Operating Procedures for Aerial Waterfowl Breeding Ground Population and Habitat Surveys in North America, Section III, revised 1987." Any deviation from these procedures or changes in aerial or ground coverage is explained in detail.

WEATHER AND HABITAT CONDITIONS: This section of the report expands on the Mid-May Habitat Conditions Report and updates these comments. Reference is made to Table 6 (Exhibit 6) which contains weather data from representative weather stations within the reporting unit from the previous summer through the survey period. The results of the adjusted pond counts compared to the previous year, the previous 10-year mean, and the long-term mean as presented in Table 4 (Exhibit 4) are also discussed in this section of the prairie unit reports.

BREEDING POPULATION ESTIMATES: Waterfowl breeding population status based upon the estimates contained in Tables 1 and 2 (Exhibits 1 and 2) is discussed in detail, and comparisons are made between the current year and the previous year, the previous 10-year mean, and the long-term mean estimates. Suspected reasons for any changes are discussed.

LONE DRAKE INDEX: The current year's figures, as well as all previous years of record, are presented in Table 3 (Exhibit 3) and their implications in interpreting the nesting season chronology and survey timing are discussed in the prairie unit reports.

CONCLUSIONS: Based on weather, habitat conditions, breeding population estimates, and other available information, the anticipated recruitment within the reporting unit is discussed.

TABLES: The following titles are used for the tables included in each survey unit report. The bush unit reports do not include tables on the lone-drake index, May ponds, and air-ground visibility rates.

1. Table 1. Long-term trend in waterfowl breeding population estimates by species in (reporting unit), 19__-__. (Exhibit 1)

2. Table 2. Status of waterfowl breeding population estimates by species and stratum in (reporting unit), comparing 19__ with 19__, the 19__-19__ previous 10-year mean, and the 19__-19__ long-term mean. (Exhibit 2)
3. Table 3. Lone drake index: long-term trend expressed as a percentage of total drakes in (reporting unit), 19__-19__. (Exhibit 3)
4. Table 4. Long-term trend in May pond estimates by stratum in (reporting unit), comparing 19__ with 19__, the 19__-19__ previous 10-year mean, and the 19__-19__ long-term mean. (Exhibit 4)
5. Table 5. Air-ground visibility rate calculation for ducks, coots, and ponds for (reporting unit) in strata __-__, May 19__. (Exhibit 5)
6. Table 6. Precipitation summaries for selected reporting stations in (reporting unit), September 19__ through May 19__. (Exhibit 6)
7. Table 7. Survey design for (reporting unit), May 19__. (Exhibit 7)

One copy of the Waterfowl Breeding Population and Habitat Survey - Stratum Summary Form (Exhibit 17) for each stratum becomes a part of the survey unit report and will follow Table 7.

The Segment Data Forms and originals of all supporting Field Data Forms, Field Data Form (stratum summaries), Ground Crew Field Data Forms and Air-Ground Comparison Segment--Ground Crew Summary Forms are submitted directly to the Assistant Branch Chief-Surveys, immediately upon return of the Flyway Biologist to the field station after the survey is completed. Express Mail is used for this purpose whenever possible. Reference is made to the appendix for report addressees and distribution. This report and all supporting data are due on 10 June for prairie units and 20 June for bush units.

Table 1. Long-term trend in waterfowl breeding population estimates by species in Southern Duckland, 1955-85 (estimates in thousands)^a

Species	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985
Ducks											
Dabblers											
Mallard	2,443.1	3,026.5	2,866.8	1,911.4	2,237.2	2,138.8	1,504.6	1,944.4	1,713.7	1,412.1	1,232.9
Am. black duck	1.1	.0	.0	.0	.7	.0	.0	.0	.0	.0	.0
Gadwall	753.5	721.9	600.8	532.9	695.5	474.5	409.5	619.7	573.2	460.4	588.4
Am. wigeon	817.5	664.6	460.6	641.5	812.8	610.4	332.2	594.5	675.0	366.7	453.4
Green-winged teal	311.0	321.9	222.3	302.1	463.7	276.6	185.3	269.8	255.8	131.4	237.4
Blue-winged teal	2,622.7	1,799.6	1,631.3	902.4	1,482.8	1,307.2	1,110.1	818.2	963.2	874.9	1,327.3
N. shoveler	730.3	823.0	422.7	426.6	692.4	494.8	333.8	699.1	792.8	352.7	670.9
N. pintail	2,050.3	2,549.5	672.5	961.7	1,580.0	897.5	526.2	1,222.1	1,029.3	542.1	520.6
Subtotal	9,729.5	9,907.0	6,877.0	5,678.6	7,965.1	6,199.8	4,401.7	6,167.8	6,003.0	4,140.3	5,030.9
Divers											
Redhead	288.6	270.5	227.6	184.1	227.9	323.0	213.3	123.9	233.4	306.9	326.8
Canvasback	305.7	283.3	253.5	128.9	280.9	221.0	111.1	181.1	212.7	166.6	118.8
Scaups	622.0	495.0	702.2	526.1	796.5	629.0	277.1	498.2	844.8	549.2	328.2
Ring-necked duck	62.1	49.1	45.0	51.8	54.0	63.9	47.2	80.5	25.1	48.0	24.7
Goldeneyes	18.4	18.0	73.1	15.8	40.2	33.1	22.8	23.4	18.3	17.1	30.5
Bufflehead	62.0	33.5	52.5	58.1	61.8	34.8	40.0	33.5	52.2	38.5	36.9
Ruddy duck	114.8	97.3	122.7	68.1	134.5	77.1	76.6	301.2	149.9	88.5	163.7
Subtotal	1,473.6	1,246.7	1,476.6	1,032.9	1,595.8	1,381.9	788.1	1,241.8	1,536.4	1,214.8	1,029.6
Miscellaneous											
Oldsquaw	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Eiders	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Scoters	8.6	1.3	5.1	3.2	15.1	2.1	.6	4.0	3.0	3.7	3.2
Mergansers	4.7	3.4	2.1	4.1	8.2	4.3	9.2	2.2	3.7	11.2	1.2
Subtotal	13.3	4.7	7.2	7.3	23.3	6.4	9.8	6.2	6.7	14.9	4.4
Total ducks	11,216.4	11,158.4	8,360.8	6,718.8	9,584.2	7,588.1	5,199.6	7,415.8	7,546.1	5,370.0	6,064.9
Canada goose											
Am. coot	8.7	12.9	9.3	12.6	15.3	15.9	21.5	22.6	30.7	33.0	32.7
Subtotal	1,177.4	1,513.0	440.0	433.6	1,309.1	634.3	395.1	282.8	673.4	566.7	662.8

^aAdjusted for visibility bias.

Table 1 (continued). Long-term trend in waterfowl breeding population estimates by species in Southern Duckland, 1955-85 (estimates in thousands)^a

Species	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
Ducks										
Dabblers										
Mallard	1,104.1	1,965.6	2,015.8	2,119.9	2,164.1	2,938.3	3,401.5	2,670.9	2,367.0	2,079.5
Am. black duck	.0	.0	.0	.0	.0	.0	.7	.0	.0	.0
Gadwall	506.6	677.9	469.2	689.7	556.9	524.4	637.1	572.9	334.0	626.9
Am. wigeon	774.8	845.1	690.1	584.4	787.1	812.1	914.3	673.1	1,050.6	751.2
Green-winged teal	126.6	273.0	373.6	215.7	521.0	439.2	390.0	243.8	313.0	303.7
Blue-winged teal	1,070.0	1,496.7	1,395.7	862.4	1,034.0	2,160.5	1,917.8	1,819.7	1,842.8	1,520.9
N. shoveler	308.1	843.6	793.6	485.9	750.4	760.8	907.7	922.0	538.4	705.2
N. pintail	716.6	1,595.4	1,671.0	774.4	1,956.1	2,417.2	2,222.1	2,261.6	1,006.3	2,186.1
Subtotal	4,606.8	7,697.3	7,409.0	5,732.4	7,769.6	10,052.5	10,391.2	9,164.0	7,452.1	8,173.5
Divers										
Redhead	90.5	149.5	213.1	69.2	210.6	189.3	279.6	138.1	179.2	179.0
Canvasback	111.3	112.9	148.7	91.0	197.0	255.4	191.8	137.7	229.0	181.7
Scaups	155.2	185.6	316.4	189.4	306.5	323.0	221.7	239.2	223.0	377.9
Ring-necked duck	26.7	18.5	12.0	1.7	7.5	16.1	8.3	9.8	9.6	49.1
Goldeneyes	9.9	11.8	16.2	25.4	9.8	17.5	16.0	5.1	23.3	15.7
Bufflehead	33.3	20.0	25.9	43.7	56.1	28.9	18.4	48.1	25.3	26.9
Ruddy duck	48.4	83.3	77.7	87.2	58.6	52.1	76.0	97.8	153.5	131.1
Subtotal	475.3	581.6	810.0	507.6	846.1	882.3	811.8	675.8	842.9	961.4
Miscellaneous										
Oldsquaw	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Eiders	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Scoters	10.0	3.1	2.4	7.5	4.4	.6	6.2	3.0	5.7	2.7
Mergansers	3.3	.0	.0	.6	.9	5.5	.8	.0	1.4	.3
Subtotal	13.3	3.1	2.4	8.1	5.3	6.1	7.0	3.0	7.1	3.0
Total ducks	5,095.4	8,282.0	8,221.4	6,248.1	8,621.0	10,940.9	11,210.0	9,842.8	8,302.1	9,137.9
Canada goose	2.5	3.2	1.6	1.0	5.3	4.1	8.8	10.0	6.5	8.7
Am. coot	254.4	333.3	357.9	450.7	407.5	900.7	963.1	569.7	931.9	1,088.6

^aAdjusted for visibility bias.

Table 1 (continued). Long-term trend in waterfowl breeding population estimates by species in Southern Duckland, 1955-85 (estimates in thousands)^a

Species	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964
Ducks										
Dabblers										
Mallard	3,948.4	5,339.8	4,518.6	5,066.4	2,192.1	3,287.7	1,650.4	1,133.6	1,552.5	1,379.1
Am. black duck	.8	.0	1.9	.0	.0	.0	.0	.0	.0	.0
Gadwall	366.8	436.9	390.9	144.6	185.0	237.1	717.8	632.0	307.5	408.7
Am. wigeon	997.7	1,460.4	859.4	1,113.4	571.9	598.2	453.8	760.0	505.9	488.2
Green-winged teal	430.1	574.4	316.3	238.1	142.4	245.6	412.1	199.2	207.7	201.8
Blue-winged teal	2,758.2	3,428.6	2,528.0	1,891.7	1,247.8	1,337.3	1,223.1	674.6	459.9	1,022.1
N. shoveler	809.2	1,037.3	757.4	396.9	302.2	720.6	482.5	426.8	199.2	375.3
N. pintail	4,113.3	5,105.7	2,700.5	1,556.9	645.2	1,418.7	846.6	596.7	835.2	853.4
Subtotal	13,424.4	17,383.1	12,073.0	10,408.0	5,286.6	7,845.2	5,786.3	4,422.9	4,067.9	4,728.6
Divers										
Redhead	225.8	470.8	280.2	150.9	115.1	140.3	57.2	157.6	36.3	84.5
Canvasback	240.7	362.1	330.5	234.2	76.0	107.7	99.9	166.9	87.2	121.6
Scaups	723.9	1,074.1	757.0	440.0	577.3	409.1	673.2	324.3	153.6	133.5
Ring-necked duck	31.4	19.4	12.6	13.4	31.8	15.9	9.6	.0	24.0	11.2
Goldeneyes	13.3	23.9	19.9	11.0	15.0	24.1	11.9	8.7	4.4	2.9
Bufflehead	11.1	13.4	22.2	10.1	13.7	19.4	12.4	2.9	17.4	19.3
Ruddy duck	155.8	330.2	158.1	72.4	269.4	133.1	80.9	67.3	40.3	37.6
Subtotal	1,402.0	2,293.9	1,580.5	932.0	1,098.3	849.6	945.1	727.7	363.2	410.6
Miscellaneous										
Oldsquaw	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Eiders	.0	.0	.0	.0	.0	.0	.0	.0	.0	.0
Scoters	6.5	16.2	1.0	12.0	5.8	6.6	3.1	.0	4.6	7.1
Mergansers	.3	.0	.0	.0	.3	7.2	1.7	.0	6.0	1.5
Subtotal	6.8	16.2	1.0	12.0	6.1	13.8	4.8	.0	10.6	8.6
Total ducks	14,833.3	19,693.2	13,654.5	11,352.0	6,391.0	8,708.6	6,736.2	5,150.6	4,441.7	5,147.8
Canada goose										
Am. coot	1,265.3	2,104.6	1,503.3	448.2	489.4	383.5	211.0	252.5	109.4	202.0

^aAdjusted for visibility bias.

Table 2. Status of waterfowl breeding population estimates by species and stratum in Southern Duckland, comparing 1985 with 1984, the 1975-1984 previous 10-year mean, and the 1955-1984 long-term mean (estimates in thousands).^a

Species	Stratum (1985)					1985 Total	1984 Total	10-Year Mean	Long-Term Mean	Percent Change from	
	30	31	32	33	34					1984	Long-Term Mean
Ducks											
Dabblers											
Mallard											
Am. black duck											
Gadwall											
Am. wigeon											
Green-winged teal											
Blue-winged teal											
W. shoveler											
W. pintail											
Subtotal											
Divers											
Redhead											
Canvasback											
Scaups											
Ring-necked duck											
Goldeneyes											
Bufflehead											
Ruddy duck											
Subtotal											
Miscellaneous											
Oldsquaw											
Eiders											
Scoters											
Mergansers											
Subtotal											
Total ducks											
Canada goose											
Am. coot											

(NOTE: This table provided for format only.
Data will be supplied at a later date.)

^aAdjusted for visibility bias.

Table 3. Lone drake index: long-term trend expressed as a percentage of total drakes in Southern Duckland, 1956-1985.^a

Year	Mallard	Pintail	Canvasback	Total
1956	76.44	82.68	63.21	78.46
1957	83.49	85.97	75.44	83.83
1958	79.38	81.33	73.68	79.55
1959	74.58	69.44	46.39	72.96
1960	85.92	82.47	71.57	84.65
1961	73.90	69.94	44.97	71.90
1962	51.66	36.35	39.83	47.32
1963	82.81	82.92	77.77	82.59
1964	85.21	82.32	65.64	83.47
1965	82.11	83.69	68.02	81.77
1966	81.75	82.66	77.57	82.98
1967	86.80	82.13	56.50	83.80
1968	80.24	75.50	47.65	77.88
1969	88.37	85.10	64.14	85.92
1970	82.10	78.28	65.35	79.60
1971	79.33	76.58	64.68	77.62
1972	81.14	78.24	60.52	79.12
1973	78.85	74.25	42.26	75.02
1974	85.75	79.60	76.59	82.42
1975	83.06	74.88	72.78	79.08
1976	78.52	78.68	74.02	78.23
1977	69.62	62.99	41.93	65.54
1978	83.78	79.50	61.05	80.91
1979	81.30	77.87	63.50	78.85
1980	77.46	75.73	57.12	75.64
1981	75.68	71.78	46.04	73.37
1982	82.18	75.04	49.94	78.25
1983	75.04	73.80	65.77	73.97
1984	76.63	77.43	60.21	75.30
1985	78.04	73.44	78.30	77.03
10-year mean (1975-84)	78.33	74.77	59.24	75.91
Long-term mean (1956-84)	79.37	76.35	61.75	77.57

^aUnadjusted for visibility bias.

Table 4. Long-term trend in May pond estimates by stratum in Southern Duckland, comparing 1985 with 1984, the 1975-1984 previous 10-year mean, and the 1961-1984 long-term mean (estimates in thousands).

Year	Stratum						Total
	30	31	32	33	34	35	
1955 ^b	525.7	884.7	958.5	213.4	1,198.8	733.1	4,474.2
1956 ^b	315.9	414.9	729.4	103.2	746.9	452.6	2,762.8
1957 ^b	164.1	399.3	344.2	72.9	636.2	299.4	1,916.1
1958 ^b	210.0	319.0	367.8	103.2	466.8	344.1	1,815.9
1959 ^b	57.5	218.7	159.9	79.4	169.0	77.4	761.9
1960 ^b	166.2	692.6	394.8	90.0	487.4	270.3	2,101.4
1961	142.2	219.4	252.2	80.3	67.5	48.0	809.6
1962	176.8	422.8	343.1	49.8	291.3	64.8	1,348.6
1963	144.7	198.1	268.4	43.2	230.9	125.3	1,010.6
1964	203.3	369.1	333.1	66.8	493.7	403.7	1,869.7
1965	327.9	439.9	610.1	112.2	442.1	337.6	2,289.8
1966	350.8	587.3	595.1	133.0	593.7	404.9	2,664.8
1967	282.3	642.1	688.8	194.9	545.1	299.0	2,652.2
1968	219.5	312.6	383.4	61.8	123.5	58.5	1,159.3
1969	383.2	465.5	774.7	138.7	267.1	179.6	2,208.8
1970	289.6	628.7	763.8	106.8	721.3	518.1	3,028.3
1971	302.5	418.3	509.2	123.8	608.6	391.7	2,354.1
1972	356.5	659.8	364.7	64.5	552.2	306.2	2,303.9
1973	259.3	453.5	317.6	83.3	227.6	117.0	2,458.3
1974	465.6	910.9	822.0	133.8	945.4	462.1	3,739.8
1975	395.2	806.1	785.2	192.8	821.4	401.3	3,402.0
1976	201.9	398.9	553.3	96.9	657.1	671.5	2,579.6
1977	176.0	254.7	265.5	44.6	338.7	170.3	1,249.8
1978	274.0	393.5	566.1	161.7	545.4	280.7	2,221.4
1979	433.3	697.4	660.0	130.2	667.7	481.1	3,069.7
1980	265.5	311.3	358.2	48.1	273.3	137.2	1,393.6
1981	145.9	160.4	126.1	28.5	97.3	52.6	610.8
1982	284.0	630.6	705.3	119.3	251.5	213.8	2,204.5
1983	384.9	715.5	712.0	96.0	464.6	323.4	2,696.4
1984	283.1	548.4	267.0	35.2	250.6	127.0	1,511.3
1985	622.9	737.8	723.6	108.1	558.5	207.1	2,958.0
10-year mean	284.4	491.7	499.9	95.3	436.8	285.9	2,094.0
Long-term mean	281.2	485.2	501.0	97.8	436.6	274.0	2,117.4
Percent Change							
1985 from 1984	+120	+35	+171	+207	+123	+63	+96
1985 from the 10-year mean	+119	+50	+45	+13	+28	-28	+41
1985 from the long-term mean	+122	+52	+44	+11	+28	-24	+40

^aAdjusted for visibility bias.
^b1955-1960 unadjusted for visibility bias.
^cLong-term mean utilizes only the years of comparable data (adjusted for visibility bias).

Table 5. Air-ground visibility rate calculation for ducks, coots, and ponds for Southern Duckland in strata 30-33 May 1985

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--AIR-GROUND VISIBILITY RATE CALCULATION FORM

Province/State: S. Duckland			Strata: 30-33			May 6-23, 1985			Sheet 4 of 4			
Segment	Kinistino		Ceylon		Goodwater		Dummer		30-33 Total		30-33 SGP	
Species	G/A	Rate	G/A	Rate	G/A	Rate	G/A	Rate	G/A	Rate	G/A	Rate
Mallard	76		22		22		50		772			
Am. black duck	42		4		20		24		414	1.8647		
Gadwall												4.8000
Am. wigeon	14		32		6		56		344			
G.W. teal	6		2		2		16		90	3.8222		
B.W. teal	18		14		2		16		228			
N. shoveler	2						4		38	6.0000		
N. pintail	10		2						96			
Redhead	56		14		10		34		224	3.636		
Canvasback	14		4		4		6		768			
Scaups	32		4		10		18		142	6.4085		
Ring-necked duck	12		2				6		460			
Goldeneyes	14		12		6		24		162	2.8395		
Bufflehead	4		6		4		8		537			
Ruddy duck	13						28		252	2.1310		
Oldsquaw	1						12		175			
Eiders	8				4		18		23	7.6087		
Scoters	6						10		89			
Mergansers	16		2		2		131		70	1.2714		
TOTAL	15						65		303			
Am. coot									137	2.2117		
Ponds									8			3.9363
									2	4.0000		5.5000
									0	--		2.7000
									16			7.5382
									2	8.0000		6.5000
									102			6.0000
									19	5.3684		1.3000
									0	--		2.0000
									0	--		
									532			
									112	4.7500		
									2,168			
									2,098	1.0334		

Form 5

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Table 6. Precipitation summaries for selected reporting stations in Southern Duckland, September 1984 through May 1985.^a

Station	9/1/84 to 10/31/84		11/1/84 to 3/31/85		4/1/85 to 5/22/85		9/1/84 to 5/22/85		Diff- erence	Percent Change
	Actual	Normal	Actual	Normal	Actual	Normal	Actual	Normal		
N. Battleford	113.8	41.5	87.4	90.6	124.4	44.0	325.6	176.1	+149.5	+84.9
Saskatoon	116.8	49.1	65.8	87.9	113.6	45.1	296.2	182.1	+114.1	+62.7
Prince Albert	147.8	61.1	105.3	89.6	90.1	47.9	343.2	198.6	+144.6	+72.8
Hudson Bay	153.8	79.3	90.8	113.2	41.8	51.0	286.4	243.5	+ 42.9	+17.6
Wynyard	153.6	64.4	89.2	100.9	87.4	48.8	330.2	214.1	+116.1	+54.2
Kindersley	91.7	40.4	59.1	79.8	92.0	48.7	242.8	168.9	+ 73.9	+43.8
Swift Current	81.5	52.2	83.3	94.2	40.0	54.1	204.8	200.5	+ 4.3	+ 2.1
Moose Jaw	87.8	54.1	81.3	89.7	67.8	60.0	236.9	203.8	+ 33.1	+16.2
Yorkton	155.1	69.4	116.0	109.9	53.2	49.7	324.3	229.0	+ 95.3	+41.6
Regina	83.6	55.6	73.1	81.0	84.6	49.8	241.3	186.4	+ 54.9	+29.5
Broadview	146.0	75.5	80.9	76.8	43.2	66.5	270.1	218.8	+ 51.3	+23.4
Estevan	79.0	65.4	59.2	91.5	39.7	69.6	177.9	226.5	- 48.6	-21.5

^a Data obtained from Environment Canada (all precipitation in millimeters).

Table 7. Survey design for Southern Duckland, May 1985.

	Stratum					Total
	30	31	32	33	34	35
<u>Survey design</u>						
Square miles in stratum	18,570	21,086	37,911	11,345	13,164	9,044
Square miles in sample-waterfowl	162	144	571.5	90	175.5	126
Square miles in sample-ponds	81	72	285.75	45	87.75	63
Linear miles in sample	648	576	2,286	360	702	504
Number of transects in sample	4	5	14	6	5	6
Number of segments in sample	36	32	127	20	39	28
Expansion factor	114.630	146.431	66.336	126.056	75.009	71.778
						87.565
<u>Current year coverage</u>						
Square miles in sample-waterfowl	162	144	571.5	90	175.5	126
Square miles in sample-ponds	81	72	285.75	45	87.75	63
Linear miles in sample	648	576	2,286	360	702	504
Number of transects in sample	4	5	14	6	5	6
Number of segments in sample	36	32	127	20	39	28
Expansion factor	114.630	146.431	66.336	126.056	75.009	71.778
						87.565

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREW FIELD DATA FORM

Province/State: S. Duckland		Stratum: 31	Transect: 17	Pilot-Observer: J. McLeod	Observer: M. P. Taylor	Sheet 1 of 1														
May 20, 1985		Wind: SE	Speed: 10	Sky: ☁																
Notes	Time & Segment	Na1	B1k	Gad	Nig	GHT	BWT	Snr	Pin	Red Can	Scp	Rng Gld Buf	Rud Old	Edr Scr	Mer	TOTAL	Coot	CE	Water	
145°F 5-15-1408 long light	1308	11/12 11/15		11/1	11/1	11/1	11/1	11/1			11/1	11/1					26	10		85 4
	3	pp				ppp	pp									5				
5-15	1320	11/12 11/15		11/1	11/1	11/1	11/1	11/1			11/1	11/1				48	27	11/1	130 8	
	4	pppppppp				pp	pp									5				
SE-15	1332	11/12 11/15		11/1	11/1	11/1	11/1	11/1			11/1	11/1				31	21	11/1	125 2	
	5	pppppp														11				
	1344	11/12 11/15		11/1	11/1	11/1	11/1	11/1								14	8		75 1	
	6	ppp																		
	1356	11/12 11/15									11/1	11/1				22	20	11/1	124 2	
	7	pppppp														5				
SE-10	1408	11/12 11/15		11/1	11/1	11/1	11/1	11/1			11/1	11/1				44	29	11/1	166 1	
	8	pppppp																		
	1420	11/12 11/15		11/1	11/1	11/1	11/1	11/1								185	15		705 18	
	9	pppppp														26				
	1432	11/12 11/15		11/1	11/1	11/1	11/1	11/1												

Form 1

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WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREW FIELD DATA FORM

Instructions for completing form:Record heading information as follows:

Time: Use Universal Coordinated (Greenwich) Time (1150, 1315, 1410, etc.)
 Wind: Use True direction from which wind is blowing (NW, E, SW, etc.)
 Speed: Provide in Statute miles per hour (5, 8, 15, etc.)
 Sky: Indicate cloud cover as clear , scattered , broken , or overcast 
 Water: For each segment, enter the natural water count in the left half of the block and the artificial water count in the right half of the block. (Note: Do not double these counts on this form.)

Record waterfowl data as follows:

Lone drakes: 1111 111 etc.

Flocked drakes: (2) (4) (3) etc.
 (Note: Five or more drakes of all species are to be treated as a group.)

Pairs: pppppp etc.

Groups:    etc.
 (Note: A hen and 2 drakes will be treated as a pair and a lone drake and a hen and 3 drakes will be treated as a pair and 2 lone drakes.)

Summarize waterfowl data for each segment and for the transect using a "fractional" system as follows:

TOTAL DRAKES (Lone drakes + flocked drakes)	TOTAL PAIRS	TOTAL IN GROUPS

Province/State: N. Duckland		Stratum: Z1		Transect: 1		Pilot-Observer: R. Stamp		Observer: G. Ringneak		Sheet 1 of 1																
May 17, 1985		Wind: SW		Speed: 10		Sky: 0		Waterfowl Species																		
Notes	Time & Segment	Mal	Blk	Gad	Wig	GWT	BWT	Str	Pin	Red	Can	Scp	Rng	6ld	Buf	Rud	Old	Edr	Scr	Mer	TOTAL	Coot	CG	WF	Swn	Crm
12:30	12:30-12:35	12				+						+	+								+	10	6			
9	12:35-12:40	9				+						+	+									20				
1308	1308-1313	8				+						+	+								3	3				
1317	1317-1322	7				+						+	+								14	9				
1327	1327-1332	6				+						+	+								4	14				
1337	1337-1342	5				+						+	+								2	2				
1346	1346-1351	4				+						+	+								13	50				
1356	1356-1401	3				+						+	+								8	14				
1401	1401-1406	2				+						+	+								2	85				
1411	1411-1416	1				+						+	+													
1457	1457-1462	1				+						+	+													
1467	1467-1472	1				+						+	+													
1477	1477-1482	1				+						+	+													
1487	1487-1492	1				+						+	+													
1497	1497-1502	1				+						+	+													
1507	1507-1512	1				+						+	+													
1517	1517-1522	1				+						+	+													
1527	1527-1532	1				+						+	+													
1537	1537-1542	1				+						+	+													
1547	1547-1552	1				+						+	+													
1557	1557-1562	1				+						+	+													
1567	1567-1572	1				+						+	+													
1577	1577-1582	1				+						+														

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	TOTAL DRACKES (Lone drakes + flocked drakes)	TOTAL PAIRS
TOTAL IN GROUPS		

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREW FIELD DATA FORM

Province/State: <u>S. Duckland</u>		Stratum: <u>31</u>		Transect: <u>16-20</u>		Pilot-Observer: <u>T. Macdonald</u>		Observer: <u>M. Pintail</u>		Sheet <u>1</u> of <u>1</u>																
May 20-22, 1985		Wind: <u>—</u>		Speed: <u>—</u>		Sky: <u>—</u>																				
Time Segment	Notes	Waterfowl Species										Water														
		Mat.	Blk	Gad	Wig	GWT	BWT	Shr	Pin	Red Can	Scp	Rng	Gld	Buf	Rud	Old	Edr	Scr	Mer	TOTAL	Coot	CG	Nat Art			
16		102/23 35		18/21 15	14/10 1	114 1	46/57 15	73/26 1	46/20 10	12/8 10	14/4 15	15/24 6	11		8/2 17						35	180	99	578	605	30
17		80/35 10		3/10 1	2/2 1	5/4 1	12/5 1	19/10 5	27/6 1	6/4 10	12/5 11	12/20 1	11		3/1 4/2					185	115	44	613	705	18	
18		21/13		2/3 1	15 1	8/2 1	15/5 1	9/1 1	4/1 1	4/2 6	9/13 1	2/3 1	16		2/4				83	53	10	4/3	277	40		
19		79/21 11		1/6 1	2/1 1	2/1 1	17/13 1	17/7 1	10/2 1	7/7 11	11/2 13	11/8 11	12		4/4 6/2			11	167	73	37	11	486	19		
20		32/6 10		2/4 1	6/1 1	4/1 1	13/15 1	10/5 1	10/2 1	1/2 1	5/5 1	5/5	1						85	37	8	1	220	16		
21		314/98 66		26/40 5	24/19 1	29/12 1	108/185 15	128/49 5	97/30 10	30/23 10	44/11 18	52/70 17	2/7 1		4/7 20/6			11	871	458	198	1415	2315	123		

Form 1 MEMO-WPS 4/87

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, THE BUSH--AERIAL CREW FIELD DATA FORM																										
Province/State: <i>N. Duckland</i>		Stratum: <i>Z1</i>		Transsect: <i>1-11</i>		Pilot-Observer: <i>R. Scarp</i>		Observer: <i>G. Ragnbeck</i>		Sheet <i>1</i> of <i>1</i>																
May 10 - Jan 3, 1985		Wind: <i>—</i>		Speed: <i>—</i>		Sky: <i>—</i>		Waterfowl Species																		
Notes	Time Segment	Ma1	Bik	Gad	Wig	GWT	BWT	Shr	Pin	Red	Can	Scp	Rug	Grd	Buf	Rud	01d	Edr	Scr	Mer	TOTAL	Coot	CG	WF	Swm	Ctn
	1	25/5			3/1	2/2	1/1				2/1	3/3	4/10	3/3	1/2					17	12/30	56	1			4
	2	8/1			1/4		1/1	1/1				3/5	2/11		1/2					7/6	21					
	3	12/1			1/1	1/1	1/1					1/6	1/7		4/1				3/2		22					
	4	2/6			2/1	1/1						1/4	5/5		1/1				1/2	6/3	12		1/1			
	5	7/1			1/1	1/1						1/6	2/1		3/1				1/1	14/14	22		1/1			
	6	8/12				1/1		1/1				1/1	4/1		3/4				7/7	14/18	16		3/1			
	7	10/1				1/1						1/8	13/4		2/3				4/8	6/7	36		1/2			
	8	4/1			1/1	1/2						6/3	8/2	1/1	1/1				3/4		23		1/2		1/1	
	9	2/1			1/1							9/1	1/1		1/1				1/5		14		1/2			
	10	3/4				1/2						2/2	1/1	1/1					4/3	1/2	13		1/4			
	11	3/1			4/1	4/1						3/4	2/1		1/1				1/4	2/1	20		2/1			
Form 204	Total	84/14			14/8	9/10	1/2	1/2	1/2		24/5	30/18	58/41	54/4	12/16				16/43	35/45	238	1	2/1			4/87

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--CREW AREA VISIBILITY RATES

In the PRAIRIE UNITS: Use the current year's calculated visibility rate when it falls within the ranges indicated below (unless otherwise indicated).

	Strata 26-29	Strata 30-33	Strata 34-40	Strata 41-44	Strata 45-49
Mallard	**	**	**	**	**
Am. black duck	4.8000*	4.8000*	4.8000*	4.8000*	4.8000*
Gadwall	2.5 - 7.8	2.6 - 8.6	4.2 - 18.7	1.6 - 9.0	1.6 - 9.0
Am. wigeon	2.6 - 6.5	4.4 -15.9	3.6042*	**	15.7701*
Green-winged teal	2.2 -25.4	5.1 -37.3	14.6272*	9.5255*	8.2471*
Blue-winged teal	**	**	**	**	**
N. shoveler	2.2 - 4.0	2.7 - 6.0	2.2 - 7.0	3.3 - 8.1	3.3 - 8.1
N. pintail	**	**	**	**	**
Redhead	1.8 -10.1	2.8 -10.8	2.1 -13.4	5.3645*	3.1 -16.9
Canvasback	.8 - 4.2	1.4 - 3.6	1.5 - 3.8	2.4263*	1.4 - 4.0
Scaups	1.2 - 3.2	1.6 - 5.2	1.2 - 4.3	1.2 - 7.3	1.2 - 6.1
Ring-necked duck	1.0893*	3.9363*	4.4183*	3.9608*	23.1799*
Goldeneyes	5.5000*	5.5000*	5.5000*	5.5000*	5.5000*
Bufflehead	2.7000*	2.7000*	2.7000*	2.7000*	2.7000*
Ruddy duck	6.5846*	7.5382*	16.1736*	9.0339*	15.8491
Oldsquaw	6.5000*	6.5000*	6.5000*	6.5000*	6.5000*
Eiders	3.6000*	3.6000*	3.6000*	3.6000*	3.6000*
Scoters	1.3000*	1.3000*	1.3000*	1.3000*	1.3000*
Mergansers	2.0000*	2.0000*	2.0000*	2.0000*	2.0000*
Am. coot	1.8 -16.8	3.6 -10.1	3.1 - 9.0	5.4616*	2.8 -12.0
Canada goose	1.0000*	1.0000*	1.0000*	1.0000*	1.0000*

However, when the current year's calculated visibility rate falls outside the range indicated above, use the appropriate long-term average provided below. Use the current year's calculated visibility rate for any species not listed above.

	Strata 26-29	Strata 30-33	Strata 34-40	Strata 41-44	Strata 45-49
Gadwall	4.4964	4.8694	9.3101	4.7773	5.3770
Am. wigeon	4.5033	8.6823	3.6042*	**	15.7701*
Green-winged teal	10.1474	18.7305	14.6272*	9.5255*	8.2471*
N. shoveler	3.2494	3.8656	4.6615	3.8377	5.6642
Redhead	4.5530	6.1675	4.7040	5.3645*	7.1900
Canvasback	2.5336	2.5069	2.6341	2.4263*	2.4848
Scaups	2.3755	3.7688	2.4407	2.6882	2.9933
Am. coot	4.5281	7.4026	5.6839	5.4616*	6.7119

In the BUSH UNITS: Always use the following species visibility rates.

Mallard	3.1625	N. pintail	3.8748	Oldsquaw	6.5000
Am. black duck	4.8000	Redhead	5.3645	Eiders	3.6000
Gadwall	4.7773	Canvasback	2.4263	Scoters	1.3000
Am. wigeon	5.3988	Scaups	2.6882	Ruddy duck	9.0399
Green-winged teal	9.5255	Ring-necked duck	3.9608	Mergansers	2.0000
Blue-winged teal	10.4047	Goldeneyes	5.5000	Am. coot	5.4616
N. shoveler	3.8377	Bufflehead	2.7000	Canada goose	1.0000

* Always use these visibility rates

** Use whatever calculated

Note: This table will be updated at approximate 5-year intervals.

Province/State:	S. Duckland	Stratum:	31	Transect:	Kristina	Pilot-Observer:	T. Mallard	Observer:	M. Pintail	Sheet <u>1</u> of <u>1</u>
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MBMO-WPS 4/8

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, AIR-GROUND COMPARISON SEGMENT--GROUND CREW FIELD DATA FORM

Province/State: <i>S. Duckland</i>		Stratum: <i>31</i>		Transect: <i>Knistino</i>		May <i>23</i>		Sheet <i>1</i> of <i>31</i>												
Crew Leader: <i>W. Teal</i>		Assistants: <i>G. Rungnick & S. Coof</i>																		
Pond No.	Pond Type	Wetlands			Waterfowl Species												TOTAL	CG	Coot	
		Water Margin	Basin	Impaction (%/type)	Mal	Blk	Gad Wtg	GNT BWT	Shr Pin	Red Can	Scp Rng	Gld	Buf Rud	Old	Edm Scr	Mer				
1	5	3	8	5F 50G 20H 20I 40C 70F 80F	②													215	1	2
1A	3	1	3	100C 100C 70S																
1B	1	1	0	100C 100C 100F																
1C	3	1	5	50G 50H 50I 100F																
2	DD	1	-	5F 100BS 100F																
3	5	3	7	30C 100F														11		
3A	3	2	4	100BS 100BS 100BS														11		
4	3	2	3	100C 100S																
5	5	4	5	100A 100F 30W 70S	1P													42		1
6	4	4	5	70C 70S																
6A	1	2	0	100C 100S																
7	5	4	8	100A 70C 30F	②	P												212	P	5
8	5	4	7	70H 30C 100C														11		
9	SH	5	5	50BS 50W 50C														11		
10	5	5	10	60C 90C 10F	1													311		
11	3	3	5	100C 100S																
Dry 4	I III IV	5 5 1	6	Str. 1	Art. 1	11	11	11	11	11	11	11	11	11	11	11	11	919	11	9

Form 3

MBND-NPS 4/87

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, AIR-GROUND COMPARISON SEGMENT--GROUND CREW FIELD DATA FORM

Instructions for completing form:

Record Wetland types as follows:

Natural:

- Type 1 - TEMPORARY WETLANDS: Shallow water, small wet areas in stubble or plowed fields, and wet depressions having no moist soil vegetation which can be expected to last less than 3 weeks under normal conditions of temperature and precipitation with less than 6 inches (15 cm) of water depth. To minimize subjectivity it is stressed that water depth is the primary criterion.
- Type 3 - SEASONAL WETLANDS: Hay meadows or basins containing natural aquatics which normally can be expected to be dry by mid-summer, but retain water for at least 3 weeks after observation. These wetlands normally have a uniform vegetative cover and are 6 inches (15 cm) or more in depth. It should be pointed out that under dry fall conditions, many such ponds are plowed and refill with water in the spring. In such cases, no vegetation will be present; however, these wetlands should be classed as Type 3 if they have sufficient depth and lasting qualities.
- Type 4 - SEMI-PERMANENT WETLANDS: Ponds of sufficient depth having water that will likely persist through the brood season but may go dry during late August or early September. These wetlands will retain water during at least 7 out of 10 years. Their vegetative cover is normally clumped and may cover all but the center of the wetland.
- Type 5 - PERMANENT WETLANDS: Ponds, deep marshes, or lakes of sufficient depth that water will persist throughout the summer and fall. These wetlands are characterized by a peripheral rim of aquatic vegetation bordering an open water body.
- Stream/River: Wet = STR Dry = DST

Artificial:

Type	Wet	Dry	Type	Wet	Dry
Dugout	D	DD	Gravel Pit	GP	DGP
Flooded Dugout	FD	..	Reservoir	R	DR
Stock Dam	S	DS	Irrigation Canal	IC	DIC
Borrow Pit	BP	DUP			

Record wetland water level stage as follows:

For use with the following water level stages, a basin is the total physical depression capable of holding water, while a wetland is the portion of the basin which contains any wetland vegetation.

- Stage 1 - DRY - surface water has disappeared completely due to seepage, evaporation, or drainage.
- Stage 2 - VESTIGIAL - water occurs only as small pools or puddles within the central vegetative zone of the wetland and can be expected to dry up within a matter of days.
- Stage 3 - RECESSIONAL - water levels have receded within the central zone but still cover a fairly extensive area. In Open Water type of wetlands there is a DRAWDOWN MUDFLAT between the water edge and the emergent vegetation. In Emergent Deep marshes and Shallow marshes, a fairly wide band of the central zone is DRY.
- Stage 4 - INTERMEDIATE - in Open Deep marshes and Shallow Open Water wetlands, the central open water zone is COMPLETELY FLOODED and the water extends into the inner edge of the emergent vegetation. In Emergent Deep marshes and Shallow marshes, the water extends throughout most of the central zone.
- Stage 5 - FULL - the wetland is filled to the outer limit of the wet meadow zone.
- Stage 6 - FLOODED - water has spilled out of the wetland proper and floods the upland vegetative communities in the basin.
- Stage 7 - OVERFLOWING - water is at full supply level (FSL) of the basin and is spilling over. FSL can vary greatly from basin to basin. In some basins it is at the level of the outer edge of the wet meadow zone and in others FSL is so high that it will never be attained. (Terminal wetlands are an example of this latter situation.)
- Stage 8 - DUGOUT-RECESSIONAL - water has receded well below the rim of the dugout.
- Stage 9 - DUGOUT-FULL - water level is at or close to the rim of the dugout.

Record wetland impactions and upland conditions and use as follows:

Impactions should be recorded in appropriate space on field form by letter code and percent of area affected. Uplands should be recorded by use and percentage of use contiguous with the margin.

Basin - center to outer edge of wet meadow zone:

B = burned (A = autumn or S = spring) F = filled H = hayed
C = cultivated G = grazed I = impacted by construction
D = drained

Margin - up to 33 feet (10 m) outward from outer edge of wet meadow zone:

B = burned (A = autumn or S = spring) CL = cleared H = hayed
C = cultivated G = grazed I = impacted by construction

Upland - area contiguous with margin:

C = crop H = hayland S = stubble
F = fallow P = pasture W = woodland
G = grassland

Record waterfowl data as follows:

Waterfowl data will be recorded to the right of the pond on which they occur. Waterfowl in flight and otherwise not on a specific pond when observed will be recorded to the right of the pond most closely associated with that bird.

Lone drakes: +++++ 111 etc.

Lone hens: hhhhh etc.

Flocked drakes: (2)(4)(3) etc. (5 or more drakes of any species are recorded as a group)

Pairs: P P P P P P P etc.

Groups: (7)(5)(9) etc. (a hen and 2 drakes will be recorded as a pair and a lone drake and a hen and 3 drakes will be recorded as a pair and 2 lone drakes)

Summarize waterfowl data for each pond and at the bottom of each page as follows:

LONE DRACKS/FLOCKED DRACKS	TOTAL PAIRS
TOTAL LONE BIRDS	TOTAL BIRDS IN GROUPS
(redhead, scaup, ring-necked duck and ruddy duck)	

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY, AIR-GROUND COMPARISON SEGMENT--GROUND CREW SUMMARY FORM

Province/State: <u>S. Duck land</u>	Stratum: <u>31</u>	Segment: <u>Kinistno</u>	May <u>23</u> , 198 <u>3</u>	Start Time: <u>7:05</u>
Crew Leader: <u>W. Teal</u>	Assistants: <u>B. Wilson, A. Shovel</u>			End Time: <u>12:45</u>
Wind Direction: <u>N</u>	Wind Speed: <u>10</u>	Cloud Cover: <u>15%</u>	Precipitation: <u>None</u>	

Species	Lone Drakes	Lone Hens	Flocked Drakes	Pairs	Grouped Birds	Total Indicated Birds
Mallard	13		15	10		76
Am. black duck						
Gadwall				4	6	14
Am. wigeon				6	6	18
Green-winged teal				5		10
Blue-winged teal				20	16	56
N. shoveler	5		3	8		32
N. Pintail	1		3	3		14
Redhead	1/	1		6		13
Canvasback			2	2		8
Scaups	1	1	2	3	6	16
Ring-necked duck	1/					
Goldeneyes						
Bufflehead						
Ruddy duck	1/					
Oldsquaw						
Eiders						
Scoters						
Mergansers						
TOTAL	39	2	48	134	34	257
Am. coot					77	77

Ponds (I) w/water	20
Ponds (III-V) w/water	204
Ponds (I,III-V) w/water	224
Artificial water areas	8
Streams/Rivers	2
Wetlands w/water	234
Dry ponds	42
Potential ponds	276
Linear miles	18
Square miles	4.5
Ponds (III-V)/mi ²	45.3
Total ponds ^{2/}	214

¹In the case of redhead, scaups, ring-necked duck, and ruddy duck, lone and flocked drakes and lone hens are multiplied by 1 (x1) in calculating total indicated birds.
²Total ponds = ponds (III-V) with water + artificial water areas + streams/rivers. Note that this figure is directly comparable to the aerial crew's water count (natural + artificial (x2)) to derive a visibility factor for water areas.

Form 4

MEMO-WES 4/87

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--AIR-GROUND COMPARISON SEGMENT - GROUND CREW SUMMARY FORM

Instructions for completing form:

Wind Direction - direction from which wind is blowing (NW, E, SW, etc.)

Wind Speed - provide speed in miles per hour (5, 8, 15, etc.)

Cloud Cover - indicate as percent of sky covered by cloud (5%, 15%, 90%, etc.)

Precipitation - indicate type (rain, snow, sleet, etc.) and relative severity (light, moderate, heavy, etc.)

Lone drakes - the total number of lone males observed.

Lone hens - the total number of lone hens observed (redhead, scaups, ring-necked duck, and ruddy duck only).

Flocked drakes - the total number of males observed with one or more other males (no females present). For the purpose of this form, flocked drakes of all species numbering 5 or more per grouping will be treated as a "group" (multiplied by 1 in calculating "total indicated birds").

Pairs - the total number of male and female groupings observed.

Grouped birds - the total number of individuals, both male and female, observed in company which cannot be separated into singles and pairs at the time of observation. All counts will be entered under this column.

Total indicated birds - the total number of birds observed on the air-ground transect with allowances for hens not observed with single and flocked drakes. Total indicated birds is calculated by multiplying lone drakes by 2, lone hens by 2, flocked drakes by 2, the number of pairs by 2, and grouped birds by 1. In the case of redhead, scaups, ring-necked duck, and ruddy duck, lone and flocked drakes and lone hens are multiplied by 1 in determining total indicated birds.

Note: Five or more drakes of all species are to be treated as a group (x1); 4 or less drakes of all species are to be treated as flocked drakes (x2).

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY -
SOUTHERN MANITOBA CREW ARE ADJUSTED MALLARD VISIBILITY
RATE CALCULATION

A general regression equation has been developed to express the positive relationship existing in strata 30 through 40 between the mallard visibility rates (mallards observed by the ground crew divided by mallards observed by the aerial crew) and pond densities as determined by ground counts.

The equation is $Y = a + 0.0236X$

where Y = the unadjusted mallard visibility rate for the combined strata in a given year.

and X = the pond density as measured by Type III-V ponds, streams and artificial water bodies counted by the ground crew on all air-ground transects within strata 34-40.

To determine the adjusted mallard visibility rates by stratum:

- (1) Substitute these data in the above equation and solve for "a."
- (2) Then rewrite the equation with the new value for "a."
- (3) Substitute the appropriate stratum's adjusted pond density (Type III-V ponds, streams and artificial water bodies counted by the aerial crew times the pond visibility rate) in the resulting equation.
- (4) Solve for Y (adjusted mallard visibility rate for that stratum).

(1)

(2)

(3)

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--STRATUM SUMMARY FORM

Stratum: 31 Dates: 5/20/85 thru 5/22/85

Species	Drakes	Pairs	Grouped Birds	Total Indicated Birds (T)	Visibility Rate (V)	Population Index (P)
Mallard	314	98	66	890	1.8647	243,014
Am. black duck						
Gadwall	26	40	5	137	3.8222	76,677
Am. wigeon	24	19		86	6.0000	75,558
Green-winged teal	20	12		64	4.3636	40,894
Blue-winged teal	103	85	15	391	5.4085	309,660
N. shoveler	128	49	5	359	2.8395	149,268
N. pintail	97	30	10	264	2.1510	82,379
Redhead	30	23	10	86	7.6087	95,817
Canvasback	46	11	39	153	1.2714	28,484
Scaups	52	20	17	209	2.2117	67,687
Ring-necked duck	2	7		16	3.9363	9,222
Goldeneyes	4			8	5.5000	6,443
Bufflehead	4	7		22	2.7000	8,648
Ruddy duck	20	6	7	39	7.5382	43,049
Oldsquaw						
Eiders						
Scoters						
Mergansers	1	1		4	2.0000	1,171
Canada goose	16	15		62	1.0000	9,079
Am. coot			198	198	4.7500	137,718

Number of observed ponds (x2)	
Pond Index (x 1.0334)	4,876
Square miles in the stratum (A)	737,843
Square miles in the sample (S)	21,086
Number of segments	144
Expansion Factor	82
	146,431

Computation of the Population Index	
P = Population Index	
A = Square miles in the stratum	
T = Total Indicated Birds	
S = Square miles in the sample	
V = Visibility Ratio	
P = A · T/S · V	

^aDrakes not doubled in calculating total indicated birds (T).
^bLong-term visibility rates from SOP used.

MBNO-WPS 4/87

Form 6

WATERFOAL BREEDING POPULATION AND HABITAT SURVEY--STRATUM SUMMARY FORM

Instructions for completing form:

Drakes - the total number of lone and flocked males observed.

Pairs - the total number of male and female groupings observed.

Grouped birds - the total number of individuals, both male and female, observed in company which cannot be separated into singles and pairs at the time of observation. All coots will be entered under this column.

Total indicated birds - the total number of birds observed on the air-ground segment with allowances for hens not observed with single and flocked drakes. Total indicated birds is calculated by multiplying lone drakes by 2, lone hens by 2, flocked drakes by 2, the number of pairs by 2, and grouped birds by 1. In the case of redhead, scaups, ring-necked duck, and ruddy duck, lone and flocked drakes and lone hens are multiplied by 1 in determining total indicated birds.

Note: Five or more drakes of all species are to be treated as a group (x1); 4 or less drakes of all species are to be treated as flocked drakes (x2).

Visibility ratio - the ratio of total indicated birds found by the ground crew to the total indicated birds found by the aerial crew on all air-ground segments within the applicable strata.

LONE DRAKE INDEX CALCULATION

Stratum	A Total Drakes	B Total Pairs	C Pairs and Drakes	D % Lone Drakes	E Drake Index	F Lone Drake Index	G % Lone Drakes
<u>Mallard</u>							
30	493	110	603	.8176	93,214	76,212	
31	314	98	412	.7621	65,162	49,660	
32	89	31	120	.7417	7,960	5,904	
33	83	34	117	.7094	15,064	10,686	
34	538	157	695	.7741	52,919	40,965	
35	264	84	348	.7586	25,445	19,303	
Total					259,764	202,730	.7804
<u>N. pintail</u>							
30	139	45	184	.7554	26,973	20,375	
31	97	30	127	.7638	19,329	14,763	
32	44	12	56	.7857	3,715	2,919	
33	43	24	67	.6418	9,517	6,108	
34	169	77	246	.6870	18,452	12,677	
35	97	27	124	.7823	9,080	7,103	
Total					87,066	63,945	.7344
<u>Canvasback</u>							
30	91	19	110	.8273	18,368	15,196	
31	46	11	57	.8070	11,202	9,046	
32	1	0	1	1.0000	66	66	
33	1	3	4	.2500	819	205	
34	86	33	119	.7227	9,489	6,858	
35	16	6	22	.7273	1,579	1,148	
Total					41,523	32,513	.7830
<u>Survey Area Total</u>							
30	723	174	897	.8060	138,555	111,675	
31	457	139	596	.7668	95,693	73,377	
32	134	43	177	.7571	11,741	8,889	
33	127	61	188	.6755	25,400	17,158	
34	793	267	1,060	.7481	80,860	60,491	
35	377	117	494	.7632	36,104	27,555	
Total					388,353	299,145	.7703

Instructions for completing calculations: For each stratum and species, accomplish the following mathematical process and enter the results on the appropriate lines.

1. Take A and B directly from Stratum Summary Form
2. $C = A + B$
3. $D = A \div C$
4. $E = \text{Total indicated birds (from Stratum Summary Form)} \times \text{stratum expansion factor} \div 2.$
5. $F = D \times E$

For the Survey Area Totals, add the totals in column A for each stratum. Do the same for columns B and C. Then, do the following:

1. $D = A \div C$
2. $E = \text{the sum of the stratum indices from same column}$
3. $F = \text{the sum of the stratum indices from same column}$
4. $G = F \div E$

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--
SEGMENT DATA FORM

U.S. FISH AND WILDLIFE SERVICE
OFFICE OF MIGRATORY BIRD MANAGEMENT

Pilot-Observer

J. Mallard

Observer

M. Pintail

05

Month
(1-2)

20

Day
(3-4)

85

Year
(5-6)

79

Province
(7-8)

31

Stratum
(9-10)

17

Transect
(11-12)

SPECIES	AQU No. (23-26)	03		13		04		13		05		14		06		14	
		Segment (13-14)		Time (15-16)		Segment (13-14)		Time (15-16)		Segment (13-14)		Time (15-16)		Segment (13-14)		Time (15-16)	
		Wind (17-18)	Vel. (19-20)	Sky (21-22)	Wind (17-18)	Vel. (19-20)	Sky (21-22)	Wind (17-18)	Vel. (19-20)	Sky (21-22)	Wind (17-18)	Vel. (19-20)	Sky (21-22)	Wind (17-18)	Vel. (19-20)	Sky (21-22)	
		Singles (27-30)	Pairs (31-34)	Groups (35-38)	Singles (27-30)	Pairs (31-34)	Groups (35-38)	Singles (27-30)	Pairs (31-34)	Groups (35-38)	Singles (27-30)	Pairs (31-34)	Groups (35-38)	Singles (27-30)	Pairs (31-34)	Groups (35-38)	
Mallard	1320	14	2	5	26	9	5	8	7		4	4					
Am. black duck	1330																
Gadwall	1350		1			2			2			1					
Am. wigeon	1370		1														
G.W. teal	1390	4	3		1	1											
B.W. teal	1400	4	2		1	2		1	1		3	1					
N. shoveler	1420	1			6	2		5	1	5	2	2					
N. pintail	1430				7	3		7	2		4						
Redhead	1460				4	1		1	2								
Canvasback	1470				3	3		4		6							
Scaups	1490		1			4		3	5								
Ringneck	1500																
Goldeneyes	1510																
Bufflehead	1530	3							1								
Ruddy duck	1670							2									
Oldsquaw	1540																
Eiders	1600																
Scoters	1630																
Mergansers	1290																
Whitefronts	1710																
Canada goose	1720				3			2	1								
Swans	1800																
Am. coot	2210	0			9			8			1						
Ponds (obs.x2)	9990	178			276			254			152						

Form 7

MBMO-WPS 3/87

WATERFOWL BREEDING POPULATION AND HABITAT SURVEY--SEGMENT DATA FORM

Instructions for completing form:

1. Record all data for each whole segment recorded on the field data sheet. Data from partial segment are not to be recorded on this form.
2. Use the following codes for recording indicated information:

Code No.	Month	Code No.	Wind	Code No.	Province
04	April		(Recorded on 8 points	03	Alaska
05	May		of the compass.)	04	Alberta
06	June	36	North	11	British Columbia
		18	South	40	Labrador
		09	East	43	Mackenzie
	<u>Day</u>	27	West	45	Manitoba
		04	Northeast	50	Minnesota
01	Numerical	31	Northwest	53	Montana
02	day	13	Southeast	56	New Brunswick
10	of	22	Southwest	57	Newfoundland
15	the	00	No wind	64	North Dakota
22	month			76	Quebec
etc.			<u>Velocity</u>	79	Saskatchewan
		00	Record in miles	81	South Dakota
	<u>Year</u>	10	per hour.	93	Yukon
		etc.			
67	1967				<u>Stratum, transect,</u>
68	1968		<u>Sky Cover</u>		<u>and segment</u>
69	1969				
etc.	etc.	01	Clear	01	(Record actual stratum,
		02	Scattered	05	transect, and segment
		03	Broken	20	number, using a
		04	Overcast	30	2-digit code
				etc.	

TIME: Use a 2-digit code and record to the nearest hour using the 24-hour clock based on Universal Coordinated (Greenwich) Time (Note - 14:30 Z would be 14 Z and 14:31 Z would be 15 Z).

3. For each species, record the number of drakes without hens, the number of birds in groups in appropriate columns. Example: 2 lone drakes would be 2 in the "Singles" column; 10 pairs would be recorded as 10 in the "Pairs" column; 14 birds in a group would be recorded 14 in the "Groups" column.
4. Only totals will be recorded for coot. The number of ponds observed should be doubled and the figure entered in the "Singles" column.

Note: All data recorded on a single form must reflect the same month, day, year, province, stratum, and transect. If any of these items change, prepare a separate form.

SECTION IV: WATERFOWL PRODUCTION AND HABITAT SURVEY

A. Methods

1. Survey Dates: The survey is performed during the approximate period of 1 to 21 July in the prairie crew areas and 8 to 22 July in the bush crew areas. The starting dates vary slightly each year, but ideally the surveys begin when the majority of the broods have developed beyond the Class I stage (Gollop and Marshall 1954) and are completed before Class III broods have fledged. Reconnaissance flights and ground inspections are conducted to determine the appropriate time for beginning the survey. However, for the survey to be of value in developing regulations at the Waterfowl Status Meeting set for 25 July each year, it must begin early enough to allow sufficient time to (1) sample the entire survey area, (2) analyze the data, and (3) telephone the survey results to the Assistant Branch Chief-Surveys, prior to or on 22 July.
2. Survey Flight Time: Daily survey flights begin no later than 1 hour after sunrise in prairie habitat and no later than 2 hours after sunrise in the parkland and southern bush habitats. This allows better light penetration into dense woodlands to improve visibility. In northern bush habitats, survey flights are conducted during the mid-day period due to the extended daylight in northern latitudes at this time of year. Transect flying in the prairies is completed by 12:00 noon (local standard time) each day. Because of extended daylight and logistical problems associated with subarctic conditions, the completion time for bush operations is somewhat later in the day.
3. Survey Flight Conditions: Aerial surveys are not initiated when winds consistently exceed 15 mph (13 kts), if adverse weather conditions exist (i.e. moderate to heavy rain, excessively rough air), or if visibility is poor for other reasons. Surveys are discontinued when winds exceed 25 mph (22 kts), turbulence is excessive, or if other adverse weather develops which is unsafe for flight or may compromise survey results.
4. Survey Flight Speed and Altitude: Transects are flown at ground speeds of 90-105 mph (78-90 kts). A stop watch is used to monitor aircraft ground

speed and to aid in locating segment end points. Aircraft are normally flown at 100-150 feet (30-50 m) AGL to ensure accurate identification of waterfowl species under average light conditions and to assure safe obstacle clearance. Established known-width sample transects are flown at regular intervals to maintain consistency and accuracy of the aerial crew's transect width. Reference marks are applied to the aircraft's wing struts as guides for both pilot-observer and observer to indicate the approximate transect boundaries from the appropriate altitude.

5. Survey Flight Path: Transects are centered on section lines in the prairies and along degree lines of latitude or longitude in the bush. Transects are flown as depicted in Fig. 2. The aircraft's flight path is altered only to avoid flying directly over towns, game farms, poultry, livestock, or persons. The aircraft flight path is not altered to facilitate identification of waterfowl. Transects are flown in the same direction each year as much as practical to minimize variance associated with such changes between years.
6. Survey Sample: A segment is defined as an 18-mile (29-km) sampling unit having a width of 1/8 mile (200 m), i.e. 1/16 mile (100 m) on each side of the aircraft for the waterfowl sample and a width of 1/8 mile (200 m) on the observer's side of the aircraft for the pond sample (Fig. 7). The pilot-observer in the left front seat counts all duck and coot broods and late-nesting ducks within 330 feet (100 m) on that side of the aircraft, while the observer in the right front seat counts all duck and coot broods and late-nesting ducks within 330 feet (100 m) on that side of the aircraft. The observer also counts specified types of water area (ponds) within 660 feet (200 m) on that side of the aircraft as was done on the Waterfowl Breeding Population and Habitat Survey.

A transect is a continuous series of segments. Most transects in this survey are oriented in an east-west direction and are parallel to the other transects at regular intervals within any given stratum. The survey employs a systematic random sampling of potential transect locations. The transect is the sampling unit.

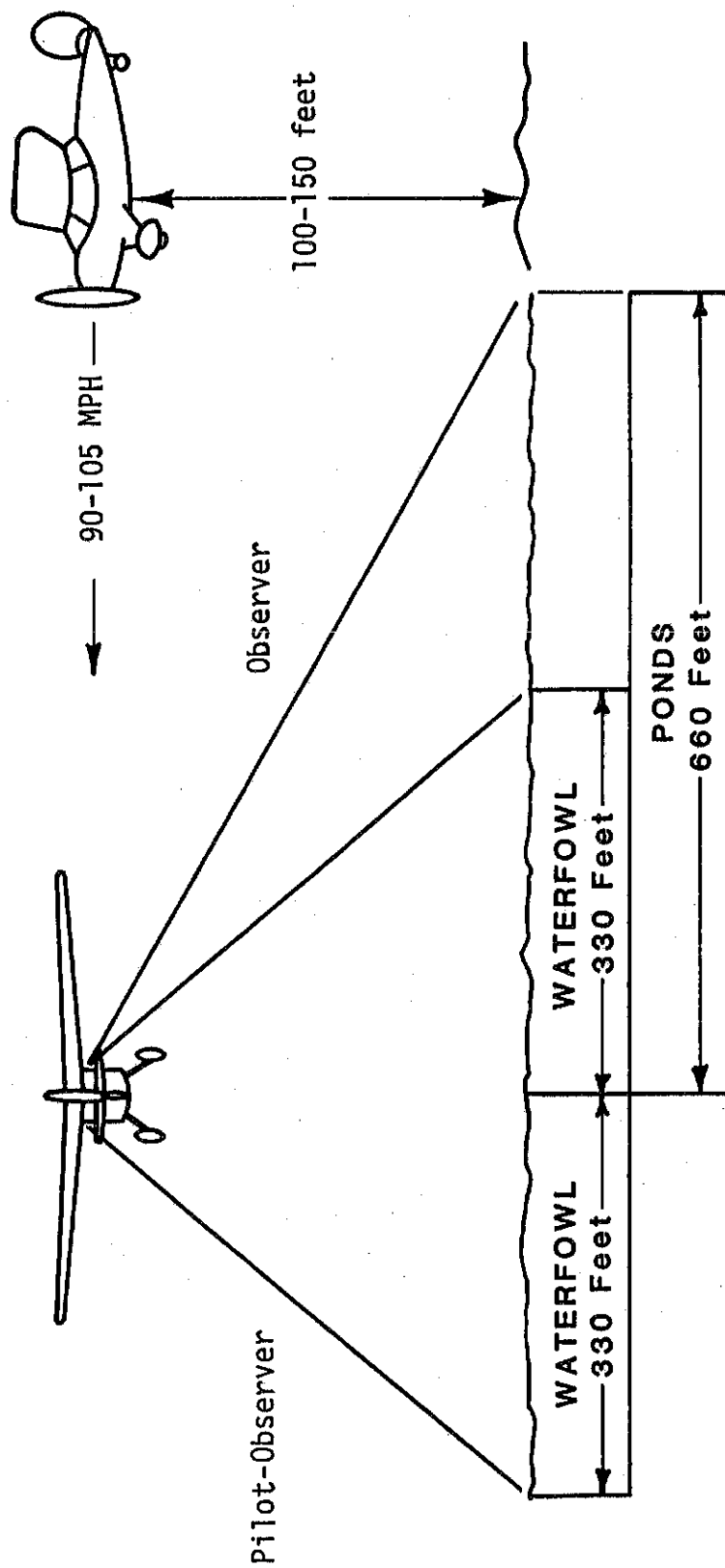


Figure 7. Survey sample for Waterfowl Production and Habitat Survey.

A stratum is a specific geographic unit encompassing areas of similar waterfowl densities and is generally of a specific habitat type. The transects usually extend from one side of a stratum to the other. Beginning near the southern boundary of a stratum, the transects extend to or near the northern boundary. Spacing between transects varies depending on waterfowl densities. Transect spacing within strata 1 through 50 varies between 14 and 60 miles (23 and 97 km).

Production surveys are not flown in Alaska, the Yukon and Northwest Territories, western Ontario, Minnesota, and Wisconsin. In the remaining 6 federal crew areas, the same transects flown during the Waterfowl Breeding Population and Habitat Survey are flown during this survey when possible. If sufficient time to cover all transects and meet the established deadline is unavailable, a representative sample of the remaining transects is selected and flown.

7. Reporting Units and Survey Crew Areas: Because it is desirable to report the data on a political unit basis and to maintain historical continuity in the various survey units, the survey data are compiled and analyzed by reporting unit (Fig. 4). Survey crew areas have been geographically organized on the basis of alignment with political boundaries, similarity in habitat, and equitable distribution of work load or flight time required to conduct the survey (Fig. 5). The strata surveyed and the strata reported on by each aerial survey crew are provided in Table 3.

For those instances where one aerial crew conducts the survey in a specified area and another crew reports on that area, it is the responsibility of the Flyway Biologist collecting the data to summarize the information and transmit it to the Flyway Biologist responsible for the report. The original and one copy of each Field Data Form and Field Data Form (stratum summary) are delivered to the Flyway Biologist responsible for that reporting unit as soon as possible following survey completion. Another copy of the Field Data Forms and Field Data Form (stratum summaries) is retained by that survey crew for referral should questions arise.

Survey area	Crew Area				Reporting Unit		
	Strata	Area (Sq. mi.)	Sample (Sq. mi.)	Linear miles	Strata	Area (Sq. mi.)	Sample (Sq. mi.)
FEDERAL							
Bush							
Northern Saskatchewan, Northern Manitoba, and the Saskatchewan River Delta	21-24	217,314	1,417.5	5,670	21-25	224,130	1,534.5
Bush subtotal		217,314	1,417.5	5,670		224,130	1,534.5
Prairies							
Southern Alberta	19,20,26-29	229,783	2,056.5	8,226	26-29	64,297	1,314.0
Southern Saskatchewan	30-33	88,912	967.5	3,870	30-35	111,120	1,269.0
Southern Manitoba	25,34-40	67,752	850.5	3,402	36-40	38,728	432.0
Montana	41-44	120,791	1,260.0	5,040	41-42	73,657	868.5
North and South Dakota	45-49	89,101	1,111.5	4,446	43-49	136,235	1,503.0
Prairie subtotal		596,339	6,246.0	24,984		424,037	5,386.5
Grand total		816,653	3,832.75	30,654		648,167	

Table 3. Waterfowl Production and Habitat Survey - crew area and reporting unit constants.

B. Data Collection

1. General

a. Strict adherence to the instructions and procedures set forth in this manual is required unless changes are approved by the Section Chief-WPS prior to the initiation of the surveys.

b. Survey data are recorded daily on Field Data Forms.

c. Field Data Forms must be prepared in duplicate. Copies are stored and mailed separately to reduce the chance of losing any field data.

d. The original field records are prepared neatly and legibly to facilitate auditing, duplication, and data transfer. Black ink or dark pencil (#2 or darker) is used on all forms.

e. Survey records are double-checked by the survey crew to ensure that transcriptions and mathematical calculations are accurate.

f. Instructions for completing the various forms are printed on the back side of each form and these instructions are strictly followed.

g. Originals of all Field Data Forms and Field Data Form (stratum summaries) are submitted directly to the Assistant Branch Chief-Surveys at the conclusion of the survey. Express Mail is used for this purpose whenever possible.

h. Color slides are useful in depicting habitat changes through the years, particularly as they relate to availability of surface water and specific land-use activities detrimental to waterfowl. It is the Flyway Biologist's responsibility to obtain representative aerial shots for this purpose and provide developed slides, appropriately labeled, to the Assistant Branch Chief-Surveys, as soon as possible following survey completion, again utilizing Express Mail if available.

2. Waterfowl Data: This survey is conducted to gather data on numbers of duck broods by age class, average brood size, number of coot broods, and number

of single lone drakes and pairs. No attempt is made to identify broods by species, but lone drakes and pairs are identified if possible and recorded. Air-ground comparison segments are not conducted during the Waterfowl Production and Habitat Survey. As a result, lone drakes and pairs that cannot be identified are included in the counts but are designated as unidentified. Grouped waterfowl (3 or more birds of mixed sexes) and flocks of 2 or more drakes are not counted. These counts make up what is called the late-nesting index (LNI). A lone drake and a pair are each considered to be a potential brood in the derivation of the Fall Flight Forecast. The number of coot broods is recorded; however, adult coots are not recorded. It is emphasized that only those broods and LNI within the transect boundaries on water bodies cut by the transect boundary are recorded.

For each water area encountered on the transect:

- a. First - Glance rapidly over the entire water area or that portion within the transect to determine what is there (number of broods and number of lone and paired adult ducks).
- b. Second - Age the broods as Class I, II, or III, and determine the number of lone drakes and pairs (by species if possible).
- c. Third - If there is sufficient time, count the number of ducklings in each Class II and III brood. Only complete Class II and III broods need be counted. If there is doubt as to whether the complete brood is visible, no record is made.

These data are recorded by segment on a tape recorder. A separate recorder is used by each aerial crew member. These data are then transcribed to Field Data Forms (Exhibit 25-prairie crews and Exhibit 26-bush crews) each afternoon following the survey flights. The data are then summarized by stratum using the same forms (Exhibit 27-prairie crews and Exhibit 28-bush crews). Table 1 (Exhibit 20) of the survey unit report presents the long-term trend in waterfowl brood and late-nesting indices for the current year and all previous years of record. Table 2 (Exhibit 21) presents the current year's waterfowl brood and late-nesting indices by stratum and compares the unit totals of the current year with the previous year, the previous 10-year mean, and the long-term mean.

3. Water Data: Water data are collected only in the prairie and parkland habitats. The aerial observer records the number of individual water areas within 660 feet (200 m) on that side of the aircraft for each segment. The pilot-observer does not count ponds for reason of flight safety. Water areas intersected by the transect boundary are counted. A tape recorder is used for this purpose as is done for the waterfowl data. Two mechanical counters are often used to assist the observer in this task, particularly in high pond density areas. The number of natural water areas (i.e. ponds, lakes, rivers, streams) is recorded on one counter, while the number of artificial water areas (i.e. reservoirs, stock dams, dugouts, and large irrigation ditches likely to contain water into mid-summer) is recorded on the other. Natural and artificial water areas are identified as such to monitor human effects on natural wetlands over the history of the surveys. The number of ponds for each segment is the sum of the counter totals. Table 3 (Exhibit 22) contains the long-term trend in unadjusted pond indices by stratum comparing the current year with the previous year, the previous 10-year mean, and the long-term mean.

Natural water areas to be counted from the air include types III, IV, and V wetlands (Shaw and Fredine 1956), rivers, and streams. Type III ponds (seasonal wetlands) refer to hay meadows or basins containing natural aquatics which normally are dry by midsummer but are expected without additional precipitation to retain water for at least 3 weeks following the observation. These wetlands normally have uniform vegetative cover and contain at least 6 inches (15 cm) of water. Under dry fall conditions, many ponds are plowed but refill with water in the spring. In these instances, only the current year's vegetation is present but these wetlands should be counted if they have sufficient water depth and lasting qualities. To minimize subjectivity, it is stressed that water depth is the primary criterion. Type IV ponds (semi-permanent wetlands) have sufficient water depth that they will likely persist through the brood season but may become dry during late August or September. These wetlands usually contain water during at least 7 out of 10 years, and the vegetation is normally clumped, covering all but the center of the wetland. Type V ponds (permanent wetlands) are deep marshes or lakes having sufficient water depth to persist throughout the summer and fall. These wetlands normally are characterized by a peripheral rim of aquatic vegetation

bordering an open body of water. Streams and rivers that meander through the transect are counted as separate water bodies each time they occur within the transect boundary. Dugouts that are inundated by a natural wetland are not recorded as artificial water areas, but if the dugout can be identified as a water area separately from the surrounding natural water area, both the dugout and the natural water area are counted.

The following water areas are not recorded:

- a. Type I ponds (temporary wetlands), refer to temporary water, sheet water, small wet areas in stubble or plowed fields, and wet depressions that have less than 6 inches (15 cm) of water depth and can be expected to last less than 3 weeks.
- b. Roadside or borrow ditches where the water is confined entirely to the ditch and small ditches used for local irrigation.
- c. Muskeg areas where water may glisten under a dense growth of grass. Open areas of water within muskeg are not counted.

C. Data Analysis

Field Data Forms designed for the prairie crew areas and bush crew areas are used to tabulate waterfowl and water data for each segment, transect, and stratum. Data collected for each stratum are summarized on the Waterfowl Production and Habitat Survey-Stratum Summary Form (Exhibit 29). Since we are interested in the numbers of late nests and not necessarily in the number of adult waterfowl observed, a pair and a lone drake are each given a value of one (1) in arriving at the LNI. The expanded species indices are obtained by multiplying each species total by the appropriate stratum expansion factor. Each stratum expansion factor is determined by dividing the total area (mi²) in the stratum by the total area (mi²) sampled within the stratum. A copy of the Stratum Summary Form for each stratum becomes part of the survey report and follows Table 5 (Exhibit 24).

The Flyway Biologist calls in the final data as soon as all the various report tables have been completed and double-checked. Specific information transmitted by phone includes general and unusual weather patterns since the spring survey, agricultural activities since

the spring survey, relative abundance, quality and distribution of late-nesting and brood-rearing cover, overall production, and any other information pertinent to interpreting the fall flight potential.

D. Survey Design

The survey design and the coverage of the current year's survey are presented in Table 5 (Exhibit 24). The first set of data presents the survey as it was designed by stratum including the area (mi²) in each stratum, the area (mi²) in the sample (both waterfowl and ponds), the number of linear miles in the sample, the number of transects in the sample, the number of segments in the sample, and the expansion factor. The second set of data includes the current year's survey coverage. If the current survey is incomplete or has been modified, the second set of data will reflect these changes. The narrative section of the report explains the reasons for any omissions or changes.

E. Reports

1. Mid-July Habitat Conditions Report: During mid-survey, the Flyway Biologist in charge of each crew area contacts the Assistant Branch Chief-Surveys, by telephone, and provides a report regarding general habitat conditions including weather patterns occurring since the spring survey, crop development and outlook, quality and availability of late nesting cover and brood rearing cover, impact of agricultural practices on duck production since the spring survey, and other pertinent information. This verbal report is due no later than 15 July for both prairie and bush units. There is no written report.

2. Waterfowl Production and Habitat Survey Report: The Waterfowl Production and Habitat Survey Report is prepared using the following outline and suggested text material.

COVER SHEET: The following statement is used--"The data presented in this report are preliminary and subject to further auditing. Final indices will be available from the U.S. Fish and Wildlife Service, Migratory Bird Management Office, Patuxent Wildlife Research Center, Laurel, Maryland 20708."

TITLE: Waterfowl Production and Habitat Survey for
(insert unit title)

STRATA SURVEYED: (insert strata numbers)

DATES: ____-____ July 19__

DATA SUPPLIED BY: United States Fish and Wildlife
Service

Air Crew

Pilot-Observer (insert name, affiliation,
and title)

Observer (insert name, affiliation,
and title)

ABSTRACT: A concise presentation of weather and habitat conditions, brood indices, late-nesting indices, pond indices, and a brief statement on overall waterfowl production in the reporting unit comparing the current year with the previous year, the previous 10-year mean, and the long-term mean.

METHODS: A statement similar to the following begins this section of the report: "The procedures followed in conducting this survey are contained in the Standard Operating Procedures for Aerial Waterfowl Breeding Ground Population and Habitat Surveys in North America, Section IV, revised 1987." Any deviation from these procedures or changes in aerial coverage is explained in detail.

WEATHER AND HABITAT CONDITIONS: This section of the report expands on the Mid-July Habitat Conditions Report and updates those comments. Reference is made to Table 4 (Exhibit 23) which contains weather data from representative weather stations within the reporting unit since the Waterfowl Breeding Population and Habitat Survey completion. Results of the unadjusted pond estimates compared to the previous year, the previous 10-year mean, the long-term mean, and the adjusted May pond estimates are presented in Table 3 (Exhibit 22) and are discussed in this section of the prairie unit reports.

PRODUCTION INDICES: Waterfowl production based upon the indices contained in Tables 1 and 2 (Exhibits 20 and 21) is discussed in detail, and comparisons are made between the current year, the previous year, the previous 10-year mean, and the long-term mean indices. Suspected reasons for any changes are discussed.

LATE-NESTING INDICES: The current year's data as presented in Tables 1 and 2 (Exhibits 20 and 21) and

STRATA SURVEYED: (insert strata numbers)

DATES: ____ - ____ July 19__

DATA SUPPLIED BY: United States Fish and Wildlife
Service

Air Crew

Pilot-Observer (insert name, affiliation,
and title)

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LATE-NESTING INDICES: The current year's data as presented in Tables 1 and 2 (Exhibits 20 and 21) and

their implications in interpreting the current year's anticipated production are discussed.

CONCLUSIONS: Based upon all available pertinent factors, both quantitative and qualitative, the anticipated fall flight from within the reporting unit is discussed and compared to the previous year, the previous 10-year mean, and the long-term mean.

TABLES: The following titles are used for the tables included in each survey report. The bush unit reports do not include a table on July ponds.

1. Table 1. Long-term trend in waterfowl brood and late-nesting indices by species in (reporting unit), 19__-19__. (Exhibit 20)
2. Table 2. Status of waterfowl brood and late-nesting indices in (reporting unit), comparing 19__ with 19__, the 19__-19__ previous 10-year mean, and the 19__-19__ long-term mean. (Exhibit 21)
3. Table 3. Long-term trend in July pond indices by stratum in (reporting unit), comparing 19__ with 19__, the 19__-19__ previous 10-year mean, the 19__-19__ long-term mean, and comparison of May with July ponds in 19__. (Exhibit 22)
4. Table 4. Precipitation summaries for selected reporting stations in (reporting unit), May through July 19__. (Exhibit 23)
5. Table 5. Survey design for (reporting unit), July 19__. (Exhibit 24)

One copy of the Waterfowl Production and Habitat Survey-Stratum Summary Form (Exhibit 29) for each stratum becomes a part of the survey unit report and will follow Table 5.

As indicated above, the results of the Waterfowl Production and Habitat Survey are relayed by telephone to the Assistant Branch Chief-Surveys, immediately following the completion of the survey but not later than noon (EDT) on 22 July (or as instructed). Flyway Biologists conducting surveys in Canada provide the Population Management Biologist, CWS, with copies of Tables 1, 2, 3, 4, 5, and all Stratum Summary Forms before leaving Canada.

The original of all Field Data Forms, Field Data Form (stratum summaries), and Stratum Summary Forms for each stratum are submitted directly to the Assistant Branch Chief-Surveys, immediately upon return of the Flyway Biologist to the field station after the survey is completed. Express Mail is used for this purpose whenever possible. Reference is made to the appendix for report addresses and distribution. This report and all supporting data are due on 30 July for both the prairie and bush units.

Table 1. Long-term trend in waterfowl brood and late-nesting indices by species in Southern Duckland, 1955-1985 (index in thousands).^a

Species	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985
Broods											
Duck brood index ^b	148.2	172.5	144.6	130.4	107.3	130.7	77.9	63.2	69.5	70.5	97.1
Average brood size ^b	4.7	4.5	5.3	4.7	5.3	4.6	4.2	4.8	4.5	4.6	5.3
Coot brood index ^c	45.0	46.0	24.8	28.3	34.1	34.2	12.5	14.9	15.6	21.5	34.9
Late-nesting index ^c											
Ducks											
Dabblers											20.9
Mallard											.0
Am. black duck											6.4
Gadwall											2.3
Am. wigeon											1.8
Green-winged teal											7.7
Blue-winged teal											1.6
N. shoveler											3.4
N. pintail											44.1
Subtotal											2.0
Divers											.7
Redhead											6.4
Canvasback											1.0
Scaups											.2
Ring-necked duck											.1
Goldeneyes											.7
Bufflehead											17.4
Ruddy duck											.0
Subtotal											.0
Miscellaneous											.0
Oldsquaw											.0
Eiders											.0
Scoters											.0
Mergansers											.0
Subtotal											20.9
Unidentified											82.4
Total ducks											

^aUnadjusted for visibility bias.^bFrom complete Class II and III broods observed.^cAs indicated by observed adult pairs and singles.(NOTE: This table provided for format only.
Data will be supplied at a later date.)

Table 1 (continued). Long-term trend in waterfowl brood and late-nesting indices by species in Southern Duckland, 1955-1985
(Index in thousands).^a

Species	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
Broods										
Duck brood index ^b	244.1	382.0	414.5	269.5	104.5	121.7	71.8	35.6	46.1	67.5
Average brood size ^b	6.6	6.0	6.0	4.4	4.3	4.8	4.7	5.4	5.5	5.8
Coot brood index ^c	-	70.5	217.7	21.9	6.0	15.3	4.9	1.1	-	-
Late-nesting index ^c										
Ducks										
Dabblers										
Mallard										
Am. black duck										
Gadwall										
Am. wigeon										
Green-winged teal										
Blue-winged teal										
N. shoveler										
N. pintail										
Subtotal										
Divers										
Redhead										
Canvasback										
Scaups										
Ring-necked duck										
Goldeneyes										
Bufflehead										
Ruddy duck										
Subtotal										
Miscellaneous										
Oldsquaw										
Eiders										
Scoters										
Mergansers										
Subtotal										
Unidentified										
Total ducks										

(NOTE: This table provided for format only.
Data will be supplied at a later date.)

^a Unadjusted for visibility bias.

^b From complete Class II and III broods observed.

^c As indicated by observed adult pairs and singles.

Table 1 (continued). Long-term trend in waterfowl brood and late-nesting indices by species in Southern Duckland, 1955-1985 (index in thousands).^a

Species	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964
Broods										
Duck brood index ^b	317.4	422.2	615.8	253.1	105.8	125.8	68.7	32.8	45.8	66.9
Average brood size ^b	6.0	5.6	6.0	4.3	3.7	3.8	4.4	4.9	5.4	5.7
Coot brood index ^c	21.0	81.8	254.2	21.6	5.4	14.8	6.0	1.1	5.2	9.2
Late-nesting index										
Ducks										
Dabblers										
Mallard										
Am. black duck										
Gadwall										
Am. wigeon										
Green-winged teal										
Blue-winged teal										
N. shoveler										
N. pintail										
Subtotal										
Divers										
Redhead										
Canvasback										
Scaups										
Ring-necked duck										
Goldeneyes										
Bufflehead										
Ruddy duck										
Subtotal										
Miscellaneous										
Oldsquaw										
Eiders										
Scoters										
Mergansers										
Subtotal										
Unidentified										
Total ducks										

(NOTE: This table provided format only.
Data will be supplied at a later date.)

^a Unadjusted for visibility bias.

^b From complete Class II and III broods observed.

^c As indicated by observed adult pairs and singles.

Table 2. Status of waterfowl brood and late-nesting indices by stratum in Southern Duckland, comparing 1985 with 1984, the 1975-1984 previous 10-year mean, and the 1955-1984 long-term mean (index in thousands).

Species	Stratum (1985)					1985 Total	1984 Total	10-Year Mean	Long-term Mean	Percent change from	
	30	31	32	33	34					1984	10-Year Mean
Broods											
Duck brood index ^b	29.6	19.6	23.0	.8	19.4	4.7	70.5	111.5		+ 38	- 13
Average brood size	5.9	4.5	5.3	6.0	5.1	5.5	4.6	4.7		+ 15	+ 13
Coot brood index ^c	11.7	15.8	2.3	.0	5.1	.0	21.5	27.7		+ 62	+ 26
Ducks											
Dabblers											
Mallard	2.8	3.2	4.0	.3	7.4	3.2	21.8				
Am. black duck	.0	.0	.0	.0	.0	.0	.0				
Gadwall	.9	.9	2.0	.0	2.5	.1	1.4				
Am. wigeon	.7	.3	.7	.0	.3	.3	1.5				
Green-winged teal	.9	.6	.1	.0	.2	.0	1.8				
Blue-winged teal	1.4	.6	2.1	.5	2.7	.4	15.8				
N. shoveler	.2	.0	.4	.0	.4	.6	1.6				
N. pintail	.9	.3	.7	.0	1.2	.3	3.4				
Subtotal	7.8	5.9	10.0	.8	14.7	4.9	46.1				
Divers											
Redhead	.7	.3	.7	.0	.0	.3	2.0				
Canvasback	.0	.3	.3	.0	.0	.1	1.1				
Scaups	.9	.0	1.9	.0	2.2	1.4	11.2				
Ring-necked duck	.0	.6	.0	.0	.4	.0	.0				
Goldeneyes	.2	.0	.0	.0	.0	.0	.2				
Bufflehead	.0	.0	.0	.0	.0	.1	.0				
Ruddy duck	.0	.0	.0	.0	.0	.0	.0				
Subtotal	2.8	1.5	.4	.0	1.9	.4	4.9				
Miscellaneous	4.6	2.7	3.3	.0	4.5	2.3	17.4				
Oldsquaw	.0	.0	.0	.0	.0	.0	.0				
Eiders	.0	.0	.0	.0	.0	.0	.0				
Scoters	.0	.0	.0	.0	.0	.0	.0				
Mergansers	.0	.0	.0	.0	.0	.0	.0				
Subtotal	.0	.0	.0	.0	.0	.0	.0				
Unidentified	1.6	2.0	5.7	.5	7.8	3.3	20.9				
Total ducks	14.0	20.6	19.0	1.3	27.0	10.4	82.4				
Total ducks							64.0				

(NOTE: Table 2 will be updated
with complete data at a
later date.)

^aUnadjusted for visibility bias.

^bFrom complete class II and III broods observed.

^cAs indicated by observed adult pairs and singles.

Table 3. Long-term trend in July pond indices by stratum in Southern Duckland, comparing 1985 with 1984, the 1975-1984 previous 10-year mean, the 1955-1984 long-term mean, and comparison of May with July ponds in 1985 (estimates in thousands).

Year	Stratum						Total
	30	31	32	33	34	35	
1955	138.6	332.1	374.5	120.5	668.5	449.0	2,083.2
1956	120.9	186.8	210.1	34.8	346.5	216.2	1,115.3
1957	59.0	136.8	127.6	18.9	260.8	77.4	680.5
1958	57.0	82.8	69.1	18.4	127.9	46.1	401.3
1959	40.1	95.9	123.0	31.5	155.6	74.1	520.2
1960	47.3	104.0	136.8	16.9	229.7	102.5	637.2
1961	41.0	35.6	51.1	10.3	32.8	22.4	193.2 ^b
1962	29.9	40.0	62.6	12.4	-	-	144.9 ^b
1963	93.0	97.2	227.8	41.6	177.5	89.4	726.5
1964	33.5	82.5	99.2	13.1	141.9	144.3	514.5
1965	112.4	188.7	289.1	88.7	267.5	69.5	915.9
1966	149.0	320.8	239.9	72.9	164.3	105.2	1,052.1
1967	86.4	136.5	192.6	44.6	101.1	50.2	611.4
1968	66.3	96.2	88.5	15.9	41.1	20.2	328.2
1969	125.4	151.8	357.0	63.0	86.6	159.2	943.0
1970	278.3	365.8	568.2	70.1	219.3	209.6	1,711.3
1971	159.1	277.6	335.9	41.9	171.7	91.6	1,077.8
1972	116.5	189.7	154.8	25.2	108.0	107.4	701.6
1973	153.1	442.7	145.3	21.7	103.5	52.4	918.7
1974	262.5	309.9	455.3	57.5	252.5	175.0	1,512.7
1975	216.7	299.6	391.1	69.1	282.5	281.9	1,540.9
1976	165.1	254.5	414.2	55.2	266.7	211.5	1,367.3
1977	101.6	187.4	183.0	19.9	154.1	72.2	718.2
1978	82.1	177.8	240.1	50.4	165.3	135.7	851.4
1979	159.6	230.8	274.2	46.9	169.2	155.8	1,036.5
1980	77.3	109.8	90.3	21.9	63.0	32.7	395.0
1981	75.7	87.0	96.3	22.9	52.2	29.6	363.7
1982	130.9	197.1	372.5	122.0	86.0	55.4	963.9
1983	134.8	313.9	237.5	44.1	366.3	99.1	1,195.7
1984	126.8	218.8	140.1	21.7	103.4	41.9	652.7
1985	186.2	292.9	173.8	20.9	177.5	55.8	907.1
10-year mean	127.1	207.7	243.9	47.4	170.9	111.6	908.6
Long-term mean	114.7	191.7	224.9	43.1	181.6	116.5	862.5
Percent Change 1985 from 1984	+ 47	+ 34	+ 24	- 4	+ 72	+ 33	+ 39
1985 from 10-year mean	+ 46	+ 41	- 29	- 56	+ 4	- 50	NC
1985 from long-term mean	+ 62	+ 53	- 23	- 52	- 2	- 52	+ 5
May ponds 1985 (adjusted)	622.9	737.8	723.6	108.1	558.5	207.2	2,958.0
Percent change May to July 1985	- 70	- 60	- 76	- 81	- 68	- 73	- 69

^a July ponds are unadjusted for visibility bias.

^b Incomplete coverage.

Table 4. Precipitation summaries for selected reporting stations in Southern Duckland, May through July 1985.^a

Station	5/23/85 to 7/18/85				4/1/85 to 7/18/85			
	Actual	Normal	Difference	Percent Change	Actual	Normal	Difference	Percent Change
N. Battleford	72.7	110.0	- 37.3	- 34	197.1	154.0	+ 43.1	+ 28
Saskatoon	82.2	106.7	- 24.5	- 23	195.8	151.8	+ 44.0	+ 29
Prince Albert	138.0	121.1	+ 16.9	+ 14	228.1	169.0	+ 59.1	+ 35
Hudson Bay	135.6	126.4	+ 9.2	+ 7	177.4	177.4	.0	NC
Wynyard	115.4	127.5	- 12.1	- 9	202.8	176.3	+ 26.5	+ 15
Kindersley	39.8	91.5	- 51.7	- 57	131.8	140.2	- 8.4	- 6
Swift Current	76.1	116.6	- 40.5	- 35	116.1	170.7	- 54.6	- 32
Moose Jaw	75.1	108.1	- 33.0	- 31	142.9	168.1	- 25.2	- 15
Yorkton	144.8	122.5	+ 22.3	+ 18	198.0	172.2	+ 25.8	+ 15
Regina	68.0	129.7	- 61.7	- 48	152.6	179.5	- 26.9	- 15
Broadview	99.0	100.8	- 1.8	- 2	142.2	167.3	- 25.1	- 15
Estevan	79.3	132.1	- 52.8	- 40	119.0	201.7	- 82.7	- 41

^aData obtained from Environment Canada (all precipitation in millimeters).

Table 5. Survey design for Southern Duckland, July 1985.

	Stratum					Total
	30	31	32	33	34	35
Survey design						
Square miles in stratum	18,570	21,086	37,911	11,345	13,164	9,044
Square miles in sample-waterfowl/ponds	81	72	285.75	45	87.75	63
Linear miles in sample	648	576	2,286	360	702	504
Number of transects in sample	4	5	14	6	5	6
Number of segments in sample	36	32	127	20	39	28
Expansion factor	229.259	292.861	132.672	252.111	150.017	143.556
						111,120
						634.5
						5,076
						40
						282
						175.130
Current year coverage						
Square miles in sample-waterfowl/ponds	81	72	285.75	45	87.75	63
Linear miles in sample	648	576	2,286	360	702	504
Number of transects in sample	4	5	14	6	5	6
Number of segments in sample	36	32	127	20	39	28
Expansion factor	229.259	292.861	132.672	252.111	150.017	143.556
						634.5
						5,076
						40
						282
						175.130

WATERFOWL PRODUCTION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREN FIELD DATA FORM

Province/State: <u>S. Duckland</u>		Stratum: <u>31</u>		Transect: <u>17</u>		Pilot-Observer: <u>J. McLeod</u>		Observer: <u>M. F. F. F.</u>		Sheet <u>1</u> of <u>1</u>	
July <u>17</u> , 19 <u>83</u>		Wind: <u>0</u>		Speed: <u>0</u>		Sky: <u>0</u>					
		Coot		Late-Nesting Index							
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WATERFOUL PRODUCTION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREW FIELD DATA FORM

Instructions for completing form:Record heading information as follows:

Time: Use Universal Coordinated (Greenwich) Time (1150, 1315, 1410, etc.)
 Wind: Use True direction from which wind is blowing (NW, E, SW, etc.)
 Speed: Provide in Statute miles per hour (5, 8, 15, etc.)
 Sky: Indicate cloud cover as clear ○, scattered ⊙, broken ⊕, or overcast ⊕
 Water: For each segment, enter the natural water count in the left half of the block and artificial water count in the right half of the block. (Note: Do not double these counts on this form.)

Record waterfowl data as follows:

Duck broods: When the age class is known or unknown and the number of ducklings is unknown (1441 111, etc.)
 When the age class is known and the complete number of ducklings is known (1-5, 1-7, 1-15, etc.)
 (the first number indicates a single brood and the second number indicates the number of ducklings in that brood)
 Note: Class 1 broods are recorded but the number of ducklings in age Class 1 broods are not recorded.

Coot broods: Only the number of broods are recorded. Class and number per brood are not recorded (1111 111, etc.)

Lone drakes: Isolated single drakes (1441 11, etc.)

Pairs: Isolated pair (PPPPPP, etc.)

Summarize brood data for each segment and for the transect using a "fractional" system as follows:

TOTAL NUMBER OF BROODS IN A GIVEN AGE CLASS
 TOTAL BROODS FOR WHICH A COUNT OF DUCKLINGS WAS OBTAINED--TOTAL DUCKLINGS COUNTED IN ALL THOSE BROODS $\left(\frac{15}{8-40}, \frac{6}{5-25}, \text{etc.} \right)$

The top figure is the total number of broods in the age class being totaled. The lower figure is in two parts, the first being the total broods for which a good count of ducklings was obtained, the second part being the total ducklings counted in those defined broods. For example, if the observed Class II broods were 1441 11, 1-5, 1-6, 1-4, the summary would be $\frac{10}{3-15}$.

Summarize late-nesting index data for each segment and for the transect using a "fractional" system as follows:

TOTAL NUMBER OF LONE DRACKES OBSERVED $\left(\frac{2}{5}, \frac{2}{7}, \frac{1}{3}, \text{etc.} \right)$
 TOTAL NUMBER OF PAIRS OBSERVED

Form 9

WATERFOWL PRODUCTION AND HABITAT SURVEY, THE BUSH-AERIAL CREW FIELD DATA FORM

Instructions for completing form:

Record heading information as follows:

Time: Use Universal Coordinated (Greenwich) Time (1150, 1315, 1410, etc.)
 Wind: Use True direction from which wind is blowing (NW, E, SW, etc.)
 Speed: Provide in Statute miles per hour (5, 8, 15, etc.)
 Sky: Indicate cloud cover as clear ○, scattered ⊕, broken ⊕, or overcast ⊕.

Record waterfowl data as follows:

Duck broods: When the age class is known or unknown and the number of ducklings is unknown (1111 111, etc.)
 When the age class is known and the complete number of ducklings is known (1-5, 1-7, 1-15, etc.)
 (the first number indicates a single brood and the second number indicates the number of ducklings in that brood)
 Note: Class 1 broods are recorded but the number of ducklings in age Class 1 broods are not recorded.

Coot broods: Only the number of broods are recorded. Class and number per brood are not recorded (111 111, etc.)

Lone drakes: Isolated single drakes (111 11, etc.)

Pairs: Isolated pair (ppppp, etc.)

Summarize brood data for each segment and for the transect using a "fractional" system as follows:

TOTAL NUMBER OF BROODS IN A GIVEN AGE CLASS
 TOTAL BROODS FOR WHICH A COUNT OF DUCKLINGS WAS OBTAINED--TOTAL DUCKLINGS COUNTED IN ALL THOSE BROODS $\left(\frac{15}{8-40}, \frac{6}{5-25}, \text{etc.} \right)$

The top figure is the total number of broods in the age class being totaled. The lower figure is in two parts, the first being the total broods for which a good count of ducklings was obtained, the second part being the total ducklings counted in those defined broods. For example, if the observed Class II broods were 111 11, 1-5, 1-6, 1-4, the summary would be $\frac{10}{3-15}$.

Summarize late-nesting index data for each segment and for the transect using a "fractional" system as follows:

TOTAL NUMBER OF LONE DRAKES OBSERVED $\left(\frac{3}{5}, \frac{2}{7}, \frac{1}{3}, \text{etc.} \right)$
 TOTAL NUMBER OF PAIRS OBSERVED

WATERFOWL PRODUCTION AND HABITAT SURVEY, THE PRAIRIES--AERIAL CREW FIELD DATA FORM

Province/State: <u>5. Duckland</u>		Stratum: <u>31</u>		Transect: <u>16-20</u>		Pilot-Observer: <u>J. Marland</u>		Observer: <u>M. Pickett</u>		Sheet <u>1</u> of <u>1</u>																		
July <u>17-18</u> , 1985		Wind: <u>---</u>		Speed: <u>---</u>		Sky: <u>---</u>																						
Duck Broods		Coot Broods		Late-Nesting Index																								
I	II	III	?	Ma	Blk	Ed	Wig	GNT	BWT	Shr	P	Red	Can	Scp	Rng	Gld	Buf	Rud	Old	Edr	Scr	Mer	TOTAL	UNID	CG	Nat	Art	
16	6	$\frac{16}{9-38}$	$\frac{3}{3-13}$	18			$\frac{0}{1}$	$\frac{1}{0}$				$\frac{1}{0}$				$\frac{1}{0}$		$\frac{2}{0}$					10	2		134	44	
17	5	$\frac{5}{4-17}$	$\frac{3}{3-13}$	13			$\frac{0}{1}$									$\frac{1}{0}$		$\frac{1}{1}$					2	2		200	22	
18	6	$\frac{5}{4-17}$	$\frac{1}{1-8}$	3			$\frac{1}{0}$											$\frac{1}{0}$					5			112	51	
19	5	$\frac{4}{2-8}$	$\frac{4}{1-7}$	16			$\frac{1}{0}$					$\frac{1}{0}$											4	0		202	28	
20		$\frac{2}{1-6}$	$\frac{2}{2-7}$	4																				1	0		139	18
10	22	$\frac{32}{20-86}$	$\frac{13}{10-98}$	54			$\frac{1}{2}$	$\frac{1}{0}$	$\frac{2}{0}$			$\frac{1}{0}$	$\frac{1}{0}$	$\frac{1}{0}$		$\frac{2}{0}$		$\frac{4}{1}$					21	5		837	163	
14																							8	2				

4/1/87

EXHIBIT 28

IV-27

WATERFOWL PRODUCTION AND HABITAT SURVEY, THE BUSH-AERIAL CREW FIELD DATA FORM

Province/State: <u>N. Duckland</u>		Stratum: <u>21</u>		Transect: <u>1-11</u>		Pilot-Observer: <u>R. Searp</u>		Observer: <u>G. Ruggback</u>		Sheet <u>1</u> of <u>1</u>																			
Date: <u>July 11-20, 1985</u>		Wind: <u>---</u>		Speed: <u>---</u>		Sky: <u>---</u>																							
Notes	Time & Segment	Duck Broods			Coot	Late-Nesting Index																							
		I	II	III		?	Mal	Blk	Gad	Wig	GMT	BWT	Shr	Pin	Red	Can	Scp	Rng	Gld	Buf	Rud	Old	Edr	Scr	Mer	TOTAL	UNID	CG	
	1	1	4	4-20						1	0						1	0							4				
	2	1	3	3-14		2	0										3	0							6				
	3																								1				
	4	1	1	1-4		1	0										1	0							2			1-4	
	5	1	2	2-10															1	0					1				
	6	2				1	0										0	1						1	5				
	7																	2	0						3				1-4
	8	1												1	0		1	0							2				1-4
	9																1	0							1				
	10																								0				
	11																	2	1						6				
			2	2-9		1	0																		1				
			2	2-11		1	8											0	1						1				
Form 9 Total		7	12	59	8-42	5	0										4	1							21				

MBMO-WPS 4/87

WATERFOWL PRODUCTION AND HABITAT SURVEY--STRATUM SUMMARY FORM

Instructions for completing form:

Brood Indices - calculate by multiplying observed broods in each category by the stratum expansion factor.

Average Brood Size - calculate by multiplying the complete Class II and III broods observed (number of broods and number of ducklings) by the stratum expansion factor. Then divide this expanded duckling index by the expanded brood index to determine the average brood size. To calculate the reporting unit average brood size, add the brood and duckling indices for all strata and divide.

Lone drakes - the total lone drakes observed within the stratum.

Pairs - the total pairs observed within the stratum.

Total indicated pairs - the total number of lone drakes and pairs observed within the stratum. Total indicated pairs is calculated by multiplying lone drakes by 1 and pairs by 1 as each is indicative of an actively breeding pair. For the purposes of this survey each indicated pair is important as they potentially will produce a brood at some point in time after the survey is completed.

Unidentified - either lone drakes or pairs which cannot be identified for reasons of poor light, glare, etc. which contribute equally to the total indicated pairs (total LNI).

Late Nesting Index - total indicated pairs expanded by the stratum expansion factor (unadjusted for visibility bias).

APPENDIX. REPORT ADDRESSEES AND DISTRIBUTION

<u>Addressees</u>	<u>Distribution</u>
A. <u>U.S. Fish and Wildlife Service</u>	
Regional Director (and Migratory Bird Coordinator), 500 N.E. Multnomah, Suite 1692, Portland, Oregon 97232	2
Regional Director (and Migratory Bird Coordinator), P.O. Box 1306, Albuquerque, New Mexico 87103	2
Regional Director (and Migratory Bird Coordinator), Federal Building, Fort Snelling, Twin Cities, Minnesota 55111	2
Regional Director (and Migratory Bird Coordinator), 75 Spring Street, S.W., Atlanta, Georgia 30303	2
Regional Director (and Migratory Bird Coordinator), One Gateway Center, Suite 700, Newton Corner, Massachusetts 02158	2
Regional Director (and Migratory Bird Coordinator), P.O. Box 25486, Denver Federal Center, Denver, Colorado 80225	2
Regional Director (and Migratory Bird Coordinator), 1101 East Tudor Road, Anchorage, Alaska 99503	2
Chief, Migratory Bird Management Office, Matomic Building, Room 536, Washington, D.C. 20240	1
Director, Migratory Bird & Habitat Research Laboratory, Patuxent Wildlife Research Center, Laurel, Maryland 20708	1
Branch Chief-Surveys, Patuxent Wildlife Research Center, Laurel, Maryland 20708	3
Director, Northern Prairie Wildlife Research Center, P.O. Box 2096, Jamestown, North Dakota 58402	1
Section Chief-WPS, 500 N.E. Multnomah, Suite 1692, Portland, Oregon 97232	1

Atlantic Flyway Representative, Patuxent Wildlife Research Center, Laurel, Maryland 20708	1
Mississippi Flyway Representative, 700 Cherry Street, Suite D, Columbia, Missouri 65201	1
Central Flyway Representative, 730 Simms, Room 456, Golden, Colorado 80401	1
Pacific Flyway Representative, 500 N.E. Multnomah, Suite 1692, Portland, Oregon 97232	1
Flyway Biologist, P.O. Box 2096, Jamestown, North Dakota 58402	1
Flyway Biologist, Regional Airport, 210 John Glenn Road, Suite 21 & 22, Lafayette, Louisiana 70508	2
Flyway Biologist, Patuxent Wildlife Research Center, Laurel, Maryland 20708	2
Flyway Biologist, 730 Simms, Room 456, Golden, Colorado 80401	2
Waterfowl Biologist-Waterfowl Investigations, P.O. Box 1287, Juneau, Alaska 99802	1
Wildlife Biologist, P.O. Box 1287, Juneau, Alaska 99802	1
Waterfowl Biologist, Wetlands Management Assistance Office, 1500 Capitol Avenue, Bismarck, North Dakota 58501	1

B. Canadian Wildlife Service

Director General, Environment Canada Headquarters, Ottawa, Ontario K1A 0E7 CANADA	2
Director, Western and Northern Region, 4999 - 98th Avenue, Edmonton, Alberta T6B 2X3 CANADA	2
Head, Population Management Section, Prairie Migratory Bird Research Centre, 115 Perimeter Road, Saskatoon, Saskatchewan S7N 0X4 CANADA	1
Director, Eastern Region, Environment Canada, Ottawa, Ontario K1A 0E7 CANADA	2

Librarian, Prairie Migratory Bird Research Centre, 115 Perimeter Road, Saskatoon, Saskatchewan S7N 0X4 CANADA	1
Population Management Biologist, 4999-98th Avenue, Edmonton, Alberta T6B 2X3 CANADA	1
Population Management Biologist, Prairie Migratory Bird Research Centre, 115 Perimeter Road, Saskatoon, Saskatchewan S7N 0X4 CANADA	1
Population Management Biologist, 501 University Crescent, Winnipeg, Manitoba R3T 2N6 CANADA	1
Population Management Biologist, P.O. Box 637, Yellowknife, Northwest Territories X1A 2N5 CANADA	1
 C. <u>State and Provincial Wildlife Management Agencies</u>	
Commissioner, Alaska Department of Fish and Game, Box 3-2000, Juneau, Alaska 99801	1
Waterfowl Program Coordinator, Alaska Department of Fish and Game, 333 Raspberry Road, Anchorage, Alaska 99502	1
Chief, Wildlife Management, Yukon Department of Renewable Resources, Box 2703, Whitehorse, Yukon Y1A 2L6 CANADA	1
Senior Biologist, Small Game, Yukon Department of Renewable Resources, Box 2703, Whitehorse, Yukon Y1A 2L6 CANADA	1
Waterfowl Ecologist, Northwest Territories Department of Renewable Resources, Box 1320, Yellowknife, Northwest Territories X1A 2L9 CANADA	1
Director, British Columbia Wildlife Branch, Minister of Environment, Parliament Buildings, Victoria, British Columbia V8V 1X4 CANADA	1
Bird and Endangered Species Specialist, British Columbia Wildlife Branch, Minister of Environ- ment, Parliament Buildings, Victoria, British Columbia V8V 1X4 CANADA	1

Director of Wildlife Branch, Fish and Wildlife Division, Department of Energy and Natural Resources, Main, North Petro Plaza, 9945-108 Street, Edmonton, Alberta T5K 2G6 CANADA	1
Provincial Wildlife Biologist, Department of Forestry, Lands and Wildlife, Petroleum Plaza, South Tower, 9945-108 Street, Edmonton, Alberta T5K 2C9 CANADA	1
Director, Wildlife Branch, Department of Parks, Recreation, and Culture, 3211 Albert Street, Regina, Saskatchewan S4S 5W6 CANADA	1
Provincial Waterfowl Biologist, Wildlife Branch, Department of Parks, Recreation and Culture, Box 3003, Prince Albert, Saskatchewan S6V 6G1 CANADA	1
Deputy Minister, Department of Natural Resources, 989 Century Street, Winnipeg, Manitoba R3H 0W4 CANADA	1
Provincial Waterfowl Biologist, Department of Natural Resources, 989 Century Street, Winnipeg, Manitoba R3H 0W4 CANADA	1
Director, Department of Fish, Wildlife and Parks, 1420 East Sixth Avenue, Helena, Montana 59601	1
Waterfowl Biologist, Department of Fish, Wildlife and Parks, 1125 Lake Elmo Drive, Billings, Montana 59105	1
Commissioner, Game and Fish Department, 100 North Bismarck Expressway, Bismarck, North Dakota 58501	1
Waterfowl Biologist, Game and Fish Department 100 North Bismarck Expressway, Bismarck, North Dakota 58501	1
Secretary, Department of Game, Fish and Parks, Anderson Building, Pierre, South Dakota 57501	1
Waterfowl Biologist, Department of Game, Fish and Parks, Anderson Building, Pierre, South Dakota 57501	1

Director, Minnesota Department of Natural Resources, Division of Fish and Wildlife, 600 Lafayette Road, Box 7, St. Paul, Minnesota 55146 1

Waterfowl Populations Biologist, Minnesota Department of Natural Resources, Wetland Wildlife Populations and Research Group, 102 - 23rd Street, Bemiji, Minnesota 56601 1

Director, Bureau of Wildlife Management, Wisconsin Department of Natural Resources, GEF II, WM/4, Madison, Wisconsin 53701 1

Waterfowl Biologist, Wisconsin Department of Natural Resources, 1210 North Palmatory Street, Horicon, Wisconsin 53032 1

D. Ducks Unlimited

Chief Biologist, 1190 Waverley Street, Winnipeg, Manitoba R3T 2E2 CANADA 1

Provincial Biologist, 10422-169 Street, Edmonton, Alberta T5P 3X6 CANADA 1

Provincial Biologist, 1606-4th Avenue, Regina, Saskatchewan S4P 3W7 CANADA 1

Provincial Biologist, #5, 1325 Markham Road, Winnipeg, Manitoba R3T 4J6 CANADA 1

E. Other

Aerial crew observer 1

Ground crew leader 1

Requesting agencies/individuals 1

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sample, and a width of 1/8 mile (200 m) on the observer's side of the aircraft for the pond sample (Fig. 3). The pilot-observer in the left front seat counts all identifiable ducks, geese, and coots within 660 feet (200 m) on that side of the aircraft while the observer in the right front seat counts all identifiable ducks, geese, and coots within 660 feet (200 m) on that side of the aircraft. The observer also counts specified types of water areas (ponds) within 660 feet (200 m) on that side of the aircraft. All waterfowl and pond data are recorded for each individual segment.

A transect is a continuous series of segments. Most transects in this survey are oriented in an east-west direction and are parallel to the other transects at regular intervals within any given stratum. The survey employs a systematic random sampling of potential transect locations. The transect is the sampling unit.

A stratum is a specific geographic unit encompassing areas of similar waterfowl densities and is generally of a specific habitat type. The transects usually extend from one side of a stratum to the other. Beginning near the southern boundary of a stratum, the transects extend to or near the northern boundary. Spacing between transects varies depending upon waterfowl densities. Transect spacing within strata 1 through 50 varies between 14 miles (23 km) and 60 miles (97 km).

7. Reporting Units and Survey Crew Areas: Because it is desirable to report the data on a political unit basis and to maintain historical continuity in the various survey units, the survey data are compiled and analyzed by reporting unit (Fig. 4). Survey crew areas have been geographically organized on the basis of alignment with political boundaries, similarity in habitat, and equitable distribution of work load or flight time required to conduct the survey (Fig. 5). The strata surveyed and the strata reported on by each aerial survey crew for the Waterfowl Breeding Population and Habitat Survey are provided in Table 1.

For those instances where one aerial crew conducts the survey in a specified area and another crew reports on that area, it is the responsibility of the Flyway Biologist collecting the data to summarize the information and transmit it to the Flyway Biologist responsible for the report. The original and one copy of each Field Data Form, Field Data Form (stratum

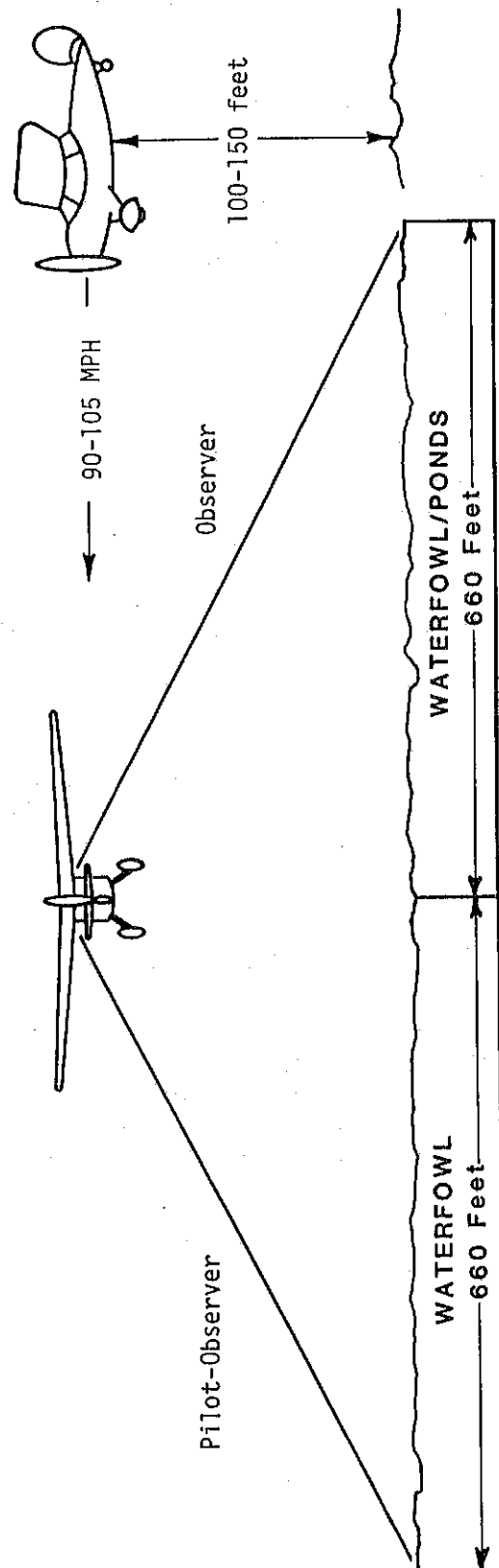


Figure 3. Survey sample for Waterfowl Breeding Population and Habitat Survey.